KeraSense: Multimodal Corneal Intelligence for Early Keratoconus Detection and Deformation Analysis



Bachelor of Technology in Artificial Intelligence &Engineering

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CERTIFICATE

This is to certify that the project entitled "KeraSense: Multimodal Corneal Intelligence for Early Keratoconus Detection and Deformation Analysis" is a bonafide work carried out by GUGGILLA DEEPU, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in [DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING] during the academic year 2026–2027

.

The work presented in this report was carried out under the guidance and supervision of the undersigned.

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DECLARATION

I hereby declare that the project entitled **“KeraSense:**
has not been submitted elsewhere for the award of any **Multimodal Corneal Intelligence for Early Keratoconus Detection and Deformation Analysis”** is my original work carried out as part of the Bachelor of Technology project under the guidance of **[Dr. R. Satya Rajendra Singh]**

The work presented in this report degree or academic qualification. All external research works, datasets, methodologies, and publications used in this project have been appropriately acknowledged.

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GUGGILLA DEEPU

Abstract

Keratoconus is a corneal disorder in which the normal shape and structure of the cornea gradually become irregular. The condition is commonly associated with corneal thinning and steepening, changes in the anterior and posterior corneal surfaces, and increasing optical irregularity. Detecting these changes at an early stage is important because patients may initially show relatively subtle abnormalities, making reliable assessment from an individual corneal map difficult. With the increasing availability of digital corneal topography and tomography, deep learning provides an opportunity to examine several complementary representations of the cornea together.

This project proposes **KeraSense**, a multimodal deep learning framework developed for the automated screening of keratoconus using seven corneal maps from the same eye. The input maps represent anterior sagittal curvature, posterior sagittal curvature, anterior elevation, posterior elevation, corneal thickness, anterior eccentricity, and posterior eccentricity. Instead of analyzing these images independently, KeraSense extracts information from all seven maps and combines their learned representations before making the final prediction.

The implemented model uses an **ImageNet-pretrained EfficientNet-B0** as a shared feature extractor. Each of the seven corneal maps is passed through the same backbone, resulting in a 1280-dimensional feature representation for each map. These seven feature vectors are then concatenated to form a 8960-dimensional multimodal feature vector. A fully connected fusion network subsequently classifies the input eye into **Keratoconus (KC)** or **Non-Keratoconus (Non-KC)**.

The dataset used in the study contains **423 development eyes and 150 independent test eyes**, with seven corresponding corneal maps available for each eye. The development data were divided into **338 training eyes and 85 validation eyes** using a stratified split, while the independent test set was kept separate for final evaluation. Before model development, the dataset was checked for incomplete map groups, duplicate maps, label inconsistencies, and identical image files between the development and independent test sets. The checks confirmed that the samples used in the experiment contained the required seven maps and that no identical image files were found across the two splits. All images were converted to a consistent RGB representation, resized to **224 × 224 pixels**, converted into tensors, and normalized using ImageNet statistics.

For the final binary experiment, class-weighted cross-entropy loss and the AdamW optimizer were used during training. Early stopping was applied to reduce unnecessary training after validation performance stopped improving. The best model checkpoint achieved a validation accuracy of **98.82%**. A probability threshold of **0.50**, selected using the validation set, was then fixed before evaluation on the independent test set.

On the untouched independent test set, KeraSense obtained an **accuracy of 93.33%**, with **82.00% sensitivity**, **99.00% specificity**, **97.62% positive predictive value**, **91.67% negative predictive value**, and an **ROC-AUC of 0.9904**. The resulting confusion matrix contained **99 true negatives, 1 false positive, 9 false negatives, and 41 true positives**. These results indicate that the model was particularly effective at identifying non-keratoconus cases while also detecting a substantial proportion of keratoconus cases.

Model interpretability was addressed using **map-wise Grad-CAM**. Grad-CAM was applied to the final convolutional feature layer of the shared EfficientNet-B0 backbone, generating a separate heatmap for each of the seven input maps. This provides a visual indication of the image regions

that influenced the model's prediction across the different corneal representations. These heatmaps are used as an interpretability aid and should not be interpreted as anatomical segmentation or direct evidence of a clinical lesion.

Overall, KeraSense demonstrates the feasibility of combining multiple complementary corneal maps within a single deep learning framework for keratoconus screening. The combination of multimodal feature fusion and map-wise visual explanation provides both a prediction and an indication of where the model obtained relevant image information. However, the present system is a **research prototype for AI-assisted screening support**, and the reported results should not be interpreted as evidence of clinical deployment or autonomous diagnosis. Further evaluation using larger, diverse, and externally collected datasets would be required before considering clinical application.

Keywords: Keratoconus, Medical Image Processing, Corneal Imaging, Corneal Topography, Multimodal Deep Learning, EfficientNet-B0, Grad-CAM, Explainable Artificial Intelligence, Deep Learning, Medical Image Classification.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CT	Corneal Thickness
CUDA	Compute Unified Device Architecture
DL	Deep Learning
EC-A	Anterior Corneal Eccentricity
EC-P	Posterior Corneal Eccentricity
Elv-A	Anterior Elevation
Elv-P	Posterior Elevation
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
KC	Keratoconus
ML	Machine Learning
MIP	Medical Image Processing
MLP	Multi-Layer Perceptron
Non-KC	Non-Keratoconus
NPV	Negative Predictive Value
PPV	Positive Predictive Value
ReLU	Rectified Linear Unit
RGB	Red, Green, Blue
ROC	Receiver Operating Characteristic
Sag-A	Anterior Sagittal Curvature
Sag-P	Posterior Sagittal Curvature
Softmax	Softmax Activation Function
SVM	Support Vector Machine
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CT	Corneal Thickness

CHAPTER 1 — INTRODUCTION

1.1 Background and Motivation

The cornea is the transparent, dome-shaped structure located at the front of the human eye. It is responsible for a major portion of the eye's optical power and plays an important role in focusing incoming light onto the retina. Since the cornea has a carefully organized shape and thickness, even relatively small structural changes can influence the quality of vision. Disorders that alter the corneal surface therefore require careful examination, particularly when the changes are progressive.

Keratoconus is one such corneal disorder in which the normally curved cornea gradually becomes thinner and develops an irregular outward protrusion. The resulting deformation changes the way light is refracted through the eye and may lead to symptoms such as blurred vision, increased astigmatism, glare, and difficulty achieving clear vision with conventional corrective lenses. The condition does not necessarily produce an obvious abnormality during its earliest stages, which makes reliable detection an important clinical challenge.

Modern ophthalmic imaging systems provide considerably more information about the cornea than a simple clinical photograph. Corneal examination can produce different representations describing surface curvature, elevation, thickness distribution, and related structural characteristics. Each representation provides a different view of the corneal condition. Consequently, examining only one image may not fully capture the combination of structural changes associated with keratoconus.

This observation provides the motivation for developing **KeraSense**, a multimodal image-analysis approach designed to examine multiple corneal representations rather than relying on a single visual pattern. The project explores the use of deep learning to identify differences among **Normal, Suspect, and Keratoconus** cases while also examining the deformation characteristics visible across the available corneal maps.

1.2 Clinical Background of Keratoconus

Keratoconus is a progressive, non-inflammatory disorder characterized by thinning and biomechanical weakening of the cornea. As the disease progresses, the corneal surface gradually loses its regular shape and develops an asymmetric, cone-like configuration. This structural alteration can cause irregular astigmatism and consequently reduce visual quality.

The clinical presentation of keratoconus can vary considerably between individuals and between different stages of the disease. In an early stage, a patient may have relatively mild visual symptoms even though subtle abnormalities are already present in the corneal structure. With progression, changes in curvature, elevation and thickness become more pronounced and may lead to increasing visual impairment.

A major difficulty in early keratoconus assessment is distinguishing a genuinely normal cornea from a cornea showing suspicious structural characteristics. This distinction is clinically important because identifying disease-related changes at an earlier stage can support appropriate monitoring and management.

Corneal imaging techniques provide quantitative and visual information that assists in this assessment. Different maps can represent characteristics such as anterior and posterior surface geometry, corneal thickness, elevation, and curvature. These measurements are complementary: one map may highlight a localized curvature abnormality, while another may

provide evidence of thinning or elevation changes. Therefore, combining information from multiple maps can provide a more comprehensive representation of corneal morphology.

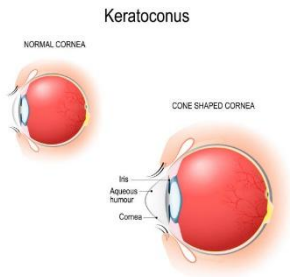


Figure 1.1 – Normal Cornea and Keratoconic Cornea

1.3 Problem Statement

Although corneal imaging provides substantial information for keratoconus assessment, interpreting multiple image representations simultaneously can be challenging. Different maps emphasize different aspects of corneal morphology, and the visual appearance of abnormality may vary considerably across patients. This makes automated classification difficult when a model is trained using only one representation of the cornea.

Another important issue is the distinction between **Normal, Suspect, and Keratoconus** cases. A binary classification system that separates healthy and diseased cases may not adequately represent the intermediate condition in which the cornea exhibits suspicious characteristics without clear evidence of established keratoconus. A useful computational system should therefore consider this intermediate category.

The problem addressed in this project is the development of an image-based deep learning approach capable of utilizing complementary corneal maps to classify cases into Normal, Suspect, and Keratoconus categories. In addition to classification, the project aims to examine whether the learned visual patterns correspond to meaningful regions associated with corneal deformation.

The central research question can therefore be stated as:

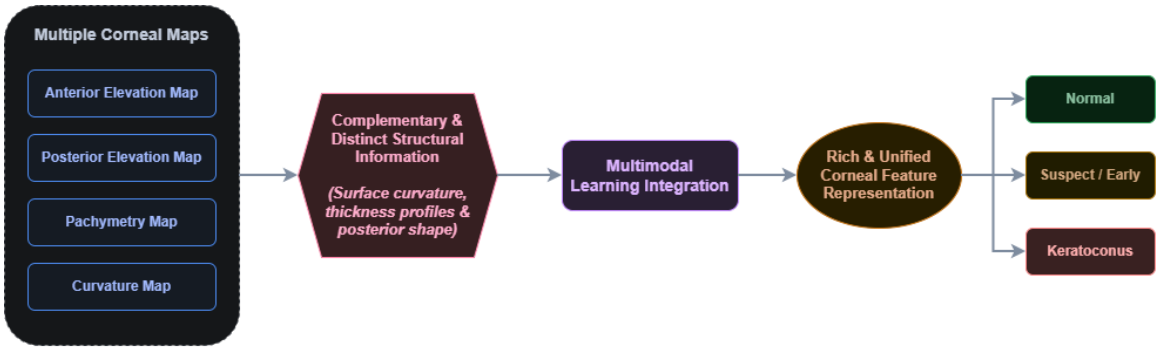


Figure 1.2: Conceptual pipeline demonstrating how integrating multiple corneal maps improves feature representation for robust keratoconus detection.

Figure 1.2 – Motivation for Multimodal Corneal Analysis

1.4 Proposed KeraSense Approach

To address the limitations of relying on a single corneal representation, the project proposes **KeraSense: Multimodal Corneal Intelligence for Keratoconus Detection**. The approach is designed around the idea that different corneal maps contain complementary visual information that can contribute to the identification of disease-related structural patterns.

The system considers multiple available corneal representations associated with each case and processes them using deep learning techniques. Instead of treating a single map as the complete description of the cornea, the multimodal approach attempts to learn a broader representation from the available information. The resulting representation is then used to distinguish between **Normal, Suspect, and Keratoconus** classes.

A convolutional neural network-based model is used as the main feature-learning component. The project employs **EfficientNet** for extracting discriminative visual features from the corneal images. EfficientNet provides a practical balance between model capacity and computational requirements, making it suitable for experimentation with medical image datasets.

The project also incorporates **Grad-CAM** to examine the regions contributing to the model's predictions. This provides an additional interpretability layer by highlighting image regions that have a stronger influence on the classification decision. Such visualization is particularly useful in medical image analysis because classification accuracy alone does not indicate whether the model is relying on meaningful structural patterns.

The overall KeraSense concept therefore combines three important aspects:

1. **Multimodal analysis** – utilization of complementary corneal representations.
2. **Deep feature learning** – automatic extraction of relevant visual characteristics.
3. **Visual interpretability** – examination of model attention using Grad-CAM.

The detailed data preparation, model architecture, training procedure, classification strategy, and evaluation methodology are presented in the subsequent chapters.

1.5 Objectives

The major objective of the project is to develop and evaluate a multimodal deep learning framework for automated analysis of corneal images associated with keratoconus.

The specific objectives are:

1. **To study the visual and structural characteristics of keratoconus** using available corneal imaging representations.
2. **To investigate the usefulness of multiple corneal maps** for representing structural changes associated with Normal, Suspect, and Keratoconus cases.
3. **To develop a deep learning-based classification system** capable of distinguishing the three target classes.
4. **To employ EfficientNet-based feature extraction** for learning discriminative representations from the input corneal images.
5. **To investigate multimodal feature integration** so that complementary information from different corneal representations can contribute to the final prediction.
6. **To evaluate the classification performance** using appropriate quantitative metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis.
7. **To improve model interpretability** using Grad-CAM visualizations and examine whether the model focuses on clinically meaningful regions.
8. **To analyze the strengths and limitations of the proposed approach** and identify possible directions for further development.

1.6 Scope of the Project

The scope of KeraSense is limited to the computational analysis of corneal imaging data for the classification of **Normal, Suspect, and Keratoconus** cases. The project focuses on image-based pattern recognition and does not attempt to replace clinical diagnosis or ophthalmological examination.

The study primarily covers the preparation of the available corneal image dataset, organization of the different image representations, development of a deep learning classification model, multimodal information integration, model evaluation, and visualization of classification-relevant regions.

The project also investigates whether combining different corneal representations provides useful information beyond that obtained from an individual image representation. This makes comparative analysis an important part of the study.

The system is intended as a **research prototype for computer-aided analysis**, rather than a clinically deployed diagnostic system. Its results are dependent on the characteristics, quality, class distribution, and labeling of the dataset used during development and evaluation.

1.7 Limitations

Despite the potential of multimodal deep learning for corneal image analysis, the proposed system has several limitations.

First, the performance of the model is strongly dependent on the dataset used for training and testing. Variations in imaging devices, acquisition protocols, patient populations, image quality, and labeling practices may affect the ability of the model to generalize to data from other clinical environments.

Second, the number of independent cases available in the dataset may be limited compared with the diversity encountered in real-world ophthalmic practice. A model trained on a relatively constrained dataset may therefore learn patterns that are specific to the dataset rather than universally applicable clinical characteristics.

Third, the **Suspect** category can be inherently difficult to classify because it represents cases with potentially abnormal characteristics that may not correspond to clearly established keratoconus. Consequently, confusion between Suspect and neighboring classes may occur.

Another limitation concerns multimodal image consistency. Different maps belonging to the same examination need to be correctly associated with one another. Missing, incorrectly paired, low-quality, or inconsistent images can affect the learned representation and ultimately influence classification performance.

Finally, Grad-CAM provides a visualization of regions that influence a model's prediction, but such visualization should not automatically be interpreted as proof of clinical reasoning. The highlighted regions require appropriate clinical interpretation and validation before the system could be considered for real-world diagnostic use.

2 LITERATURE REVIEW

2.1 Artificial Intelligence in Keratoconus Detection

The application of artificial intelligence in ophthalmology has expanded considerably with the availability of digital imaging and larger collections of clinical data. Keratoconus has received particular attention because many of its diagnostic characteristics are expressed through measurable changes in corneal shape, thickness, elevation, and curvature. These characteristics provide suitable inputs for computational models that can learn patterns associated with the disease.

Earlier computer-aided approaches commonly relied on manually selected clinical parameters obtained from corneal examinations. Such systems used measurements related to curvature, pachymetry, elevation, and other indices to distinguish normal corneas from abnormal cases. Although these approaches demonstrated that quantitative corneal information could support automated classification, their effectiveness depended strongly on the selection and quality of the input parameters.

The introduction of machine learning reduced the dependence on manually defined decision rules. Classification algorithms can identify relationships between multiple corneal measurements and diagnostic categories from previously labeled examples. This is particularly useful for keratoconus because the disease does not always produce a single characteristic abnormality. Several parameters may change simultaneously, and their relationships can be difficult to represent using fixed thresholds.

Deep learning further changed the approach by allowing models to learn image features directly from the available data. Convolutional neural networks (CNNs), in particular, have been investigated for extracting spatial patterns from corneal images and topographic representations. Instead of requiring every relevant feature to be specified beforehand, CNN-based systems can learn hierarchical representations ranging from low-level visual patterns to more complex structural characteristics.

Another important development is the recognition that keratoconus detection is not always a simple healthy-versus-diseased problem. Early or suspicious cases may contain subtle abnormalities that overlap with normal anatomical variation. Therefore, studies increasingly consider more challenging classification settings and investigate whether artificial intelligence can identify changes before the disease reaches an advanced stage.

Despite these developments, reported performance varies between studies because of differences in datasets, imaging devices, diagnostic definitions, sample sizes, and evaluation procedures. A model performing well on one dataset may not necessarily maintain the same performance when applied to images acquired under different conditions. This makes dataset-independent evaluation and careful interpretation of model results important considerations in AI-based keratoconus research.

2.2 Deep Learning for Corneal Topography and Tomography

Corneal topography and tomography provide rich representations of the geometry of the cornea. Topographic information primarily describes the surface characteristics of the cornea, while tomographic examinations can provide additional information concerning corneal thickness and the anterior and posterior surfaces. These representations are valuable for keratoconus analysis because the disease involves changes in both corneal shape and structural distribution.

Traditional analysis of these examinations often involves clinically established indices and measurements. Such measurements remain important because they provide interpretable quantitative information. However, a large image representation may contain spatial patterns that are not fully captured by a small collection of summary indices.

Deep learning provides an alternative way of analyzing these images. CNN-based models can process two-dimensional corneal maps and learn discriminative features from their spatial arrangement. In keratoconus applications, these features may correspond to localized regions of abnormal curvature, asymmetric patterns, changes in elevation, or variations in corneal thickness.

Several studies have explored different CNN architectures for the classification of corneal conditions. Transfer learning has also become common because medical imaging datasets are generally smaller than the datasets used to train large general-purpose computer vision models. A pretrained network can provide useful low-level and mid-level visual representations, after which the model can be adapted to the specific corneal classification task.

The effectiveness of deep learning depends not only on the architecture but also on how the corneal data are represented. Images generated from different examination parameters may have different visual distributions. A network trained using one type of map may therefore learn features that are highly specific to that representation. This observation motivates investigation of approaches that can make use of more than one corneal map.

Another concern is data leakage. Multiple images can originate from the same eye or examination, and randomly dividing individual images into training and testing sets may unintentionally place related samples in both subsets. Such a split can produce overly optimistic results because the model may encounter information from the same underlying case during training and evaluation. Patient-level or eye-level separation is therefore an important consideration when evaluating deep learning models for corneal imaging.

2.3 Multimap and Multimodal Corneal Analysis

The cornea cannot be completely described by a single visual representation. Different maps emphasize different physical properties, and this complementary nature provides an important opportunity for multimodal learning.

A curvature-related map may reveal changes in the distribution of corneal power, while an elevation representation can provide information about deviations of the corneal surface from a reference shape. Pachymetric information, on the other hand, describes the distribution of corneal thickness. These representations are related, but they do not contain identical information.

A model using only one map may therefore miss evidence that becomes apparent in another representation. For example, a subtle curvature abnormality may become more meaningful when considered together with a corresponding thickness or elevation pattern. Combining multiple representations can potentially provide a richer description of the corneal structure.

Multimap approaches attempt to exploit this complementary information. The different maps can be processed independently and their learned features can subsequently be combined.

Another possibility is to provide multiple representations to a shared or parallel feature-extraction network and allow the model to learn relationships between them.

The choice of fusion strategy is important. Early fusion combines information close to the input stage, whereas feature-level fusion combines learned representations after individual feature extraction. Decision-level fusion combines predictions generated by separate models. Each strategy has advantages and limitations, particularly when the input maps differ in appearance or contain different levels of diagnostic information.

Multimodal analysis also introduces additional challenges. The maps associated with one examination must be correctly matched, and missing or poor-quality inputs can affect the final prediction. Increasing the number of input representations can also increase model complexity and computational requirements.

For these reasons, multimodal learning should not simply be interpreted as adding more images to a model. The important question is whether the additional representations contribute complementary information that improves classification or provides a more reliable representation of the underlying corneal condition. This question forms an important basis for the KeraSense approach.

2.4 Deep Learning Architectures for Ophthalmic Imaging

CNNs have become a major component of medical image analysis because of their ability to learn spatial features directly from images. Common architectures such as VGG, ResNet, DenseNet, Inception, and EfficientNet have been adapted for a wide range of ophthalmic imaging tasks.

Residual networks introduced shortcut connections that help deeper models learn effectively and reduce optimization difficulties. Dense connectivity provides another strategy for feature reuse, allowing information from earlier layers to be propagated to later layers. These architectures have been used extensively in medical image classification and provide useful baselines for evaluating newer models.

EfficientNet follows a different design philosophy by jointly scaling network depth, width, and image resolution. This provides a useful balance between representational capacity and computational cost. The balance is particularly relevant for projects involving medical image datasets, where available training data and computational resources may be more limited than those used in large-scale natural image applications.

Transfer learning is another widely used strategy in ophthalmic image analysis. Rather than training a deep network completely from the beginning, a model pretrained on a large image dataset can be adapted to the target medical imaging task. The early layers generally provide useful generic visual features, while later layers can be fine-tuned to capture patterns specific to the target application.

For KeraSense, EfficientNet is considered as the primary feature-learning architecture because it provides a practical combination of feature extraction capability and computational efficiency. The architecture is not treated as a replacement for clinical knowledge; rather, it provides the computational mechanism through which patterns across corneal representations can be learned.

The comparison of architectures is also important because higher model complexity does not automatically result in better medical classification. Performance should be considered together with computational requirements, dataset size, generalization, and interpretability.

2.5 Explainable AI in Medical Image Analysis

High classification accuracy alone is not sufficient to establish the usefulness of an AI model in medical imaging. A model may achieve good numerical performance while relying on image regions that are unrelated to the clinical characteristic being studied. This creates a need for methods that can provide insight into the basis of model predictions.

Explainable artificial intelligence (XAI) refers to techniques that help interpret the behavior of machine learning models. In image-based applications, explanation methods can identify regions that have contributed strongly to a particular prediction. Such visual information can help researchers determine whether a model is responding to meaningful image characteristics or potentially learning dataset-specific artifacts.

Gradient-weighted Class Activation Mapping, commonly known as **Grad-CAM**, is one method used for this purpose. It uses gradients associated with the target prediction to generate a coarse localization map showing image regions that contribute to the model's decision. The resulting heatmap can be superimposed on the original image to provide an intuitive visual representation of model attention.

In ophthalmic imaging, visualization methods such as Grad-CAM can be useful for investigating whether learned representations correspond to recognizable structural abnormalities. For corneal images, this may help identify whether the network is concentrating on regions associated with curvature irregularity, localized deformation, or other relevant patterns.

However, explainability should be interpreted carefully. A highlighted region does not automatically demonstrate that the model has discovered a clinically validated biomarker. Explanation maps should therefore be considered as an interpretability aid rather than definitive evidence of clinical reasoning.

For KeraSense, Grad-CAM is included to complement the quantitative evaluation of the classification model. The objective is not only to determine whether the model predicts a class correctly, but also to inspect the visual regions contributing to its predictions.

Ref.	Study	Year	Dataset / Sample	Imaging Modality / Input	Model / Approach	Task	Reported Performance
[1]	Kamiya *et al.*	2019	304 KC eyes + 239 healthy eyes	Six colour-coded AS-OCT maps	Deep learning	Normal vs. KC; KC grading	Combined six-map accuracy 99.1%; posterior elevation map 99.3%
[2]	Kuo *et al.*	2020	170 KC + 28 subclinica	Corneal topography	VGG16, InceptionV3,	KC detection	ResNet152 AUC 0.995;

			l KC + 156 normal images		ResNet152		sensitivity and specificity >0.90
[3]	Al-Timemy *et al.*	2021	534 cases / 4 maps	Pentacam corneal maps	Ensemble / deep transfer learning approach	Normal vs. KC	Accuracy 98.3%
[4]	Fassbind *et al.*	2023	1,940 scans / 899 patients; independent test set: 242 scans / 188 patients	CASIA2 AS-OCT; five corneal maps	ConvNeXt-based CorNeXt	Healthy vs. abnormal; corneal pathology classification	Sensitivity 98.46%, specificity 91.96%
[5]	Systematic review of AI in KC	2024	93 original AI studies	Topography, tomography, AS-OCT and other inputs	Multiple AI approaches	KC detection, staging, progression and treatment assessment	Most studies focused on KC/subclinical KC; topography and tomography were dominant inputs
[6]	Deep learning diagnostic study using	2024	1,668 tomographic images + 611 biomecha	Pentacam tomography + Corvis ST biomechanics	Multiple CNN models	Normal, subclinical KC and KC	Best reported AUC 0.991; accuracy 0.956

	Pentacam + Corvis ST		**nical images**				
[7]	**Automated keratoconus progression triage**	**2026**	**7,396 eyes / 3,893 patients; external: 963 eyes / 519 patients**	**AS-OCT + Placido + clinical/tabular data**	**Multimodal deep learning; sequential RNN**	**Progression-risk prediction**	**Single-visit AUROC 0.84; sequential model 0.93**
[8]	**Multimodal progression classification**	**2026**	**Sirius topography + refractive data**	**Anterior/posterior corneal surface geometry + refractive data**	**Multimodal fusion**	**Four-stage KC progression classification**	**Balanced accuracy 0.852**
[9]	**Scheimpflug tomography AI study**	**2026**	**1,411 eyes; 5,602 extracted maps**	**Four Galilei corneal maps**	**DenseNet-121, ResNet-50, Inception-V3, EfficientNet-B0**	**Normal vs. KC**	**Accuracy: DenseNet 99.2%, ResNet 99.0%, Inception 98.6%, EfficientNet-B0 98.1%**

2.6 Research Gap and Motivation for KeraSense

The reviewed literature demonstrates that artificial intelligence has considerable potential for automated keratoconus analysis. Traditional machine learning approaches established the usefulness of quantitative corneal parameters, while deep learning methods demonstrated the ability to learn visual patterns directly from topographic and tomographic representations.

However, several areas remain open for further investigation.

First, a substantial portion of automated approaches focus on a **single representation or a limited set of measurements**. Since different corneal maps describe different structural properties, relying on one representation may restrict the information available to the model.

Second, although multimodal and multimap approaches address this limitation, combining several maps introduces additional questions concerning **feature fusion, input consistency, and the actual contribution of each modality**. Simply increasing the number of inputs does not guarantee improved performance.

Third, the distinction between **Normal, Suspect, and Keratoconus** remains challenging. The Suspect category is particularly important because borderline cases may contain structural patterns that overlap with both healthy and clearly keratoconic corneas. A three-class formulation can therefore provide a more informative experimental setting than a simple binary classification task.

Fourth, model interpretability remains an important consideration. A classification result without an indication of the image regions influencing that result provides limited insight into how the model reaches its decision. Incorporating explanation techniques can help investigate the visual basis of model predictions.

Based on these observations, **KeraSense** is designed around a multimodal analysis strategy that combines information from multiple corneal maps, uses deep feature extraction for classification, and incorporates Grad-CAM-based visualization for interpretation. The project specifically investigates classification of **Normal, Suspect, and Keratoconus** cases while examining whether complementary corneal representations provide useful information for the learning process.

The contribution of KeraSense is therefore positioned not simply as another CNN-based classifier, but as an attempt to bring together **multimap information, deep feature learning, three-class classification, and visual interpretability** within a single experimental framework

2.7 Comparison of Existing Approaches with KeraSense

The literature indicates a gradual transition from manually engineered corneal indices toward machine learning and deep learning systems. KeraSense is positioned at the intersection of several developments: multimodal image analysis, transfer-based deep learning, three-class classification, and explainability.

Aspect	**Conventional Approaches**	**Single-Map Deep Learning**	**Multimap Approaches**	**KeraSense**
Input information	Clinical indices / selected measurements	One corneal representation	Multiple corneal representations	**Multiple corneal maps**
Feature extraction	Handcrafted / predefined	Learned automatically	Learned automatically	**Deep learned features**
Classification	Often rule/threshold based	Binary or multiclass	Binary/multiclass	**Normal / Suspect / Keratoconus**
Multimap information	Limited	Usually unavailable	Available	**Central component**
Transfer learning	Not applicable	Common	Common	**Used with EfficientNet**
Feature fusion	Limited	Not applicable	Depends on approach	**Multimodal feature integration**

Interpretability	Clinical indices are interpretable	Often limited	Varies	**Grad-CAM visualization**
Main purpose	Clinical parameter-based assessment	Automated image classification	Combine complementary image information	**Multimodal computer-aided corneal analysis**

2.8 Summary of Literature Review

The literature reviewed in this chapter shows the increasing role of artificial intelligence in corneal image analysis and keratoconus assessment. Traditional methods established the diagnostic value of corneal curvature, elevation, thickness, and related measurements, while machine learning introduced automated classification using combinations of these parameters.

Deep learning subsequently enabled direct learning from corneal images and maps, reducing dependence on manually engineered features. CNN-based architectures and transfer learning have become useful approaches for extracting visual representations from ophthalmic data. However, the information available in a single corneal representation may not fully describe the structural characteristics associated with keratoconus.

Multimap analysis provides a potential solution by allowing complementary representations of the cornea to contribute to the prediction. At the same time, multimodal systems introduce challenges related to feature fusion, image correspondence, data quality, and model complexity.

The literature also highlights the importance of interpretability in medical image analysis. Techniques such as Grad-CAM can provide visual information about the regions contributing to model predictions, although such explanations should not be treated as direct clinical evidence.

Based on these observations, KeraSense focuses on combining multiple corneal maps with deep feature learning and Grad-CAM-based interpretation for **Normal, Suspect, and Keratoconus** classification. The following chapters describe the dataset, methodology, model development, experimental procedure, and evaluation used to investigate this approach.

CHAPTER 3 — THEORETICAL AND CLINICAL BACKGROUND

3.1 Medical Image Processing for Corneal Analysis

Medical image processing involves the acquisition, enhancement, transformation, and analysis of images obtained from biological structures. In ophthalmology, imaging techniques provide a non-invasive way of examining anatomical characteristics that may not be apparent through visual inspection alone. Corneal imaging is particularly suitable for computational analysis because the geometry and thickness of the cornea can be represented numerically and visually through different types of maps.

Corneal imaging systems can generate representations of surface curvature, elevation, thickness, and other geometric properties. These representations convert physical characteristics of the cornea into spatial patterns that can subsequently be analyzed using image-processing and machine-learning techniques. In a computational system, such images can be normalized and supplied to a learning model that identifies patterns associated with different clinical categories. Image processing is important before deep learning because medical images may contain variations caused by acquisition conditions, image resolution, scaling, illumination, or other factors. Appropriate preprocessing helps maintain useful structural information while providing inputs in a consistent form.

For KeraSense, the role of medical image processing is primarily to prepare and represent the available corneal maps in a form suitable for automated feature extraction. The project focuses on the relationship between the visual information contained in the maps and the classification of cases into Normal, Suspect, and Keratoconus categories.

3.2 Keratoconus and Corneal Structural Changes

The cornea is a transparent tissue forming the anterior surface of the eye. Its regular curvature and optical properties allow incoming light to be refracted toward the retina. Maintaining an appropriate corneal shape is therefore important for clear vision.

Keratoconus is a progressive corneal disorder characterized by thinning and deformation of the cornea. Instead of maintaining a relatively regular curved surface, the affected cornea can develop localized steepening and an asymmetric protruding configuration. These changes can produce irregular astigmatism and deterioration in visual quality.

The structural changes associated with keratoconus are not restricted to one measurable property. Alterations may occur in anterior curvature, posterior curvature, elevation, and corneal thickness. The spatial relationship between these characteristics can also provide useful diagnostic information.

This makes corneal imaging particularly valuable for computational analysis. A single numerical value may describe only one aspect of the cornea, whereas a set of maps can provide a spatial representation of several structural properties simultaneously.

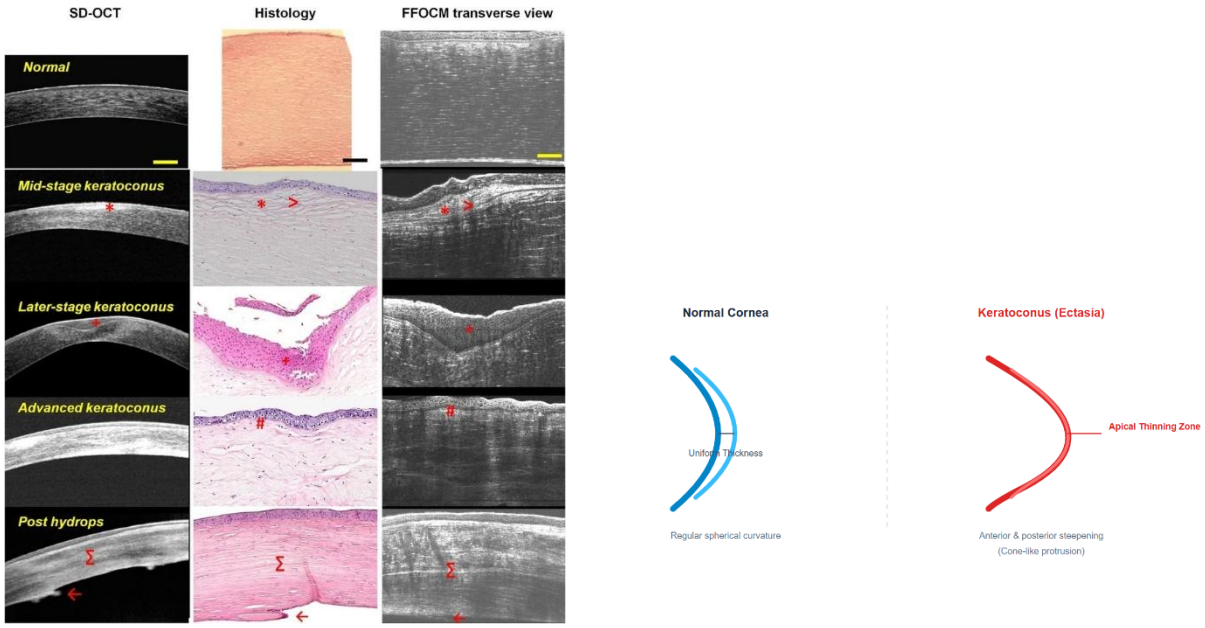


Figure 3.2: Conceptual representation of structural changes associated with keratoconus, including corneal thinning and localized deformation

3.3 Corneal Maps Used in KeraSense

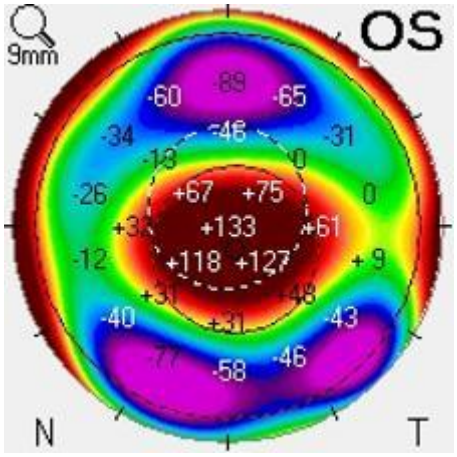
KeraSense uses seven corneal map representations to capture complementary characteristics of corneal geometry. These maps can be broadly grouped into curvature, elevation, thickness, and eccentricity representations.

Using multiple maps is important because keratoconus affects the cornea in a spatially distributed manner. A curvature map may reveal localized steepening, while elevation and thickness maps can provide additional structural information. Eccentricity-related representations provide another description of how the corneal surface differs from an idealized shape.

The seven representations considered in the project are:

Abbreviation	Representation
Sag_A	Anterior Sagittal Curvature
Sag_P	Posterior Sagittal Curvature
Elv_A	Anterior Elevation
Elv_P	Posterior Elevation
CT_A	Corneal Thickness
EC_A	Anterior Eccentricity
EC_P	Posterior Eccentricity

These maps are treated as complementary rather than interchangeable inputs. Each one provides information about a particular aspect of the corneal structure, allowing the learning system to examine a broader representation of the eye than would be possible using a single map.



3.3.1 Anterior Sagittal Curvature (Sag_A)

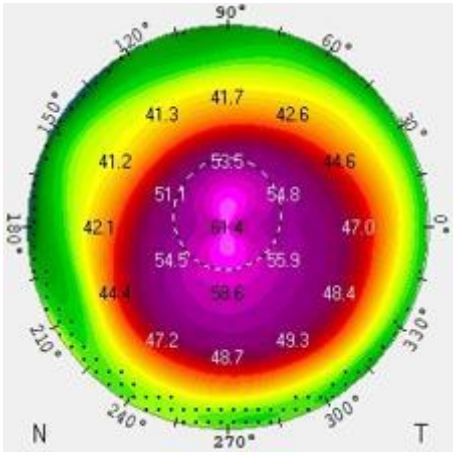
Anterior sagittal curvature represents the curvature characteristics of the anterior surface of the cornea. Curvature is related to how sharply or gradually the corneal surface changes across different locations.

In a relatively regular cornea, curvature tends to exhibit a comparatively organized spatial distribution. Keratoconus can produce localized steepening and increased asymmetry, resulting in distinctive patterns within curvature maps.

For KeraSense, the anterior sagittal curvature map provides information about the front surface of the cornea and may contain visual patterns associated with abnormal steepening.

Suggested caption for the seven-map figure:

(a) Sag_A – Anterior sagittal curvature



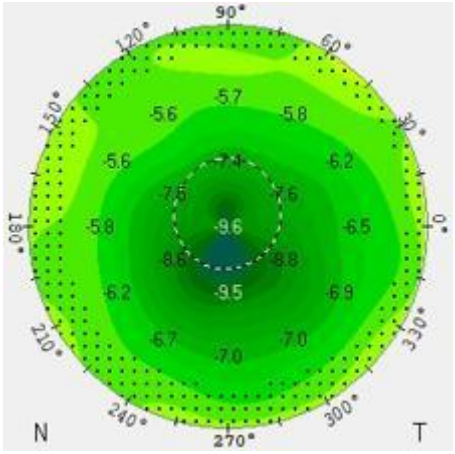
3.3.2 Posterior Sagittal Curvature (Sag_P)

Posterior sagittal curvature describes the curvature of the posterior corneal surface. Unlike the anterior surface, the posterior surface is not directly exposed to the external environment and provides additional structural information about the cornea.

Changes in posterior curvature can be useful when assessing corneal deformation because abnormalities may be detectable on the posterior surface even when changes on the anterior surface are less pronounced.

Including Sag_P alongside Sag_A therefore allows KeraSense to consider information from both major corneal surfaces.

(b) Sag_P – Posterior sagittal curvature

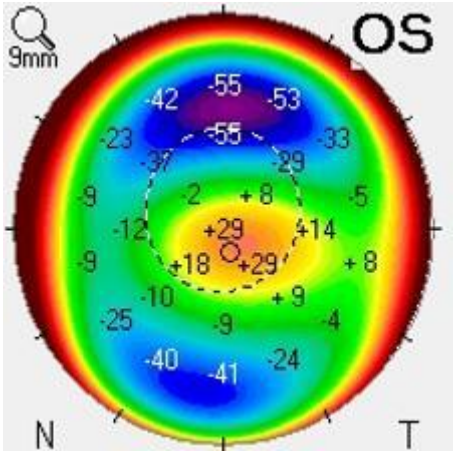


3.3.3 Anterior Elevation (Elv_A)

Anterior elevation represents the displacement of the anterior corneal surface relative to a reference surface. Instead of describing curvature directly, an elevation map indicates how the measured corneal surface differs spatially from the selected reference geometry.

Localized elevation abnormalities may contribute useful information when identifying irregular corneal morphology. When combined with curvature and thickness information, the elevation representation can provide a more complete description of structural changes.

(c) Elv_A – Anterior elevation

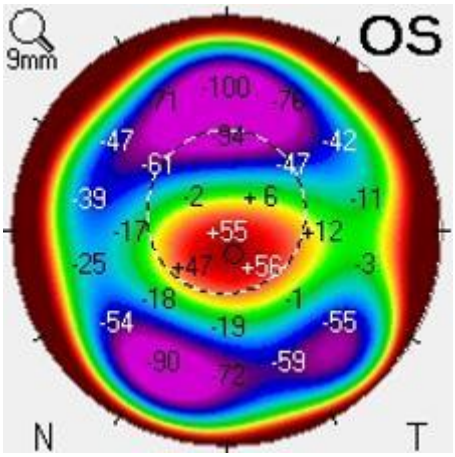


3.3.4 Posterior Elevation (Elv_P)

Posterior elevation represents the spatial displacement of the posterior corneal surface relative to a reference surface.

Posterior surface information is particularly useful in corneal assessment because structural abnormalities may involve both anterior and posterior geometry. The posterior elevation map therefore provides information that is complementary to anterior elevation and curvature maps. In KeraSense, Elv_P contributes another spatial representation through which the model can learn patterns associated with corneal deformation.

(d) Elv_P – Posterior elevation



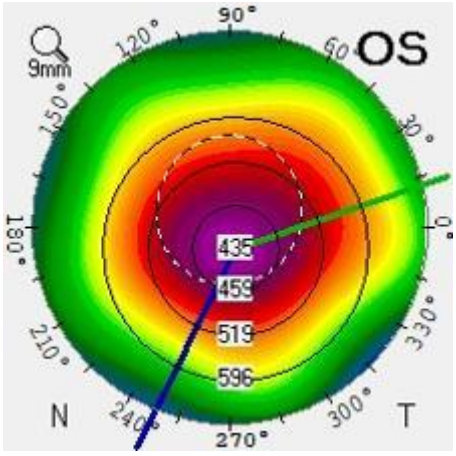
3.3.5 Corneal Thickness (CT_A)

Corneal thickness describes the thickness of the corneal tissue at different spatial locations. Unlike curvature and elevation maps, which primarily describe geometric characteristics, a thickness map provides information about the distribution of corneal tissue.

Corneal thinning is an important structural characteristic associated with keratoconus. However, the absolute thickness value alone does not completely describe the condition. The location and distribution of thickness variations can also be relevant.

For this reason, CT_A provides complementary information to the curvature and elevation maps used in KeraSense.

(e) CT_A – Corneal thickness



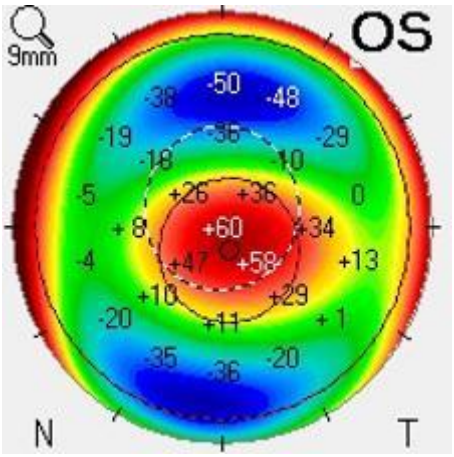
3.3.6 Anterior Eccentricity (EC_A)

Eccentricity describes the departure of a corneal surface from a simple spherical geometry. The anterior eccentricity representation therefore provides information about the shape characteristics of the anterior corneal surface.

Changes in corneal shape can modify eccentricity patterns, particularly when the surface becomes more irregular or asymmetric. This makes eccentricity another potentially useful representation for distinguishing different structural conditions.

In KeraSense, EC_A is included alongside curvature and elevation maps to provide additional geometric information about the anterior surface.

(f) EC_A – Anterior eccentricity



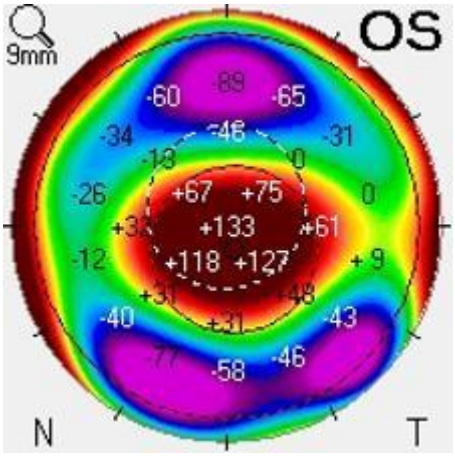
3.3.7 Posterior Eccentricity (EC_P)

Posterior eccentricity describes the shape characteristics of the posterior corneal surface relative to an idealized geometric representation.

Like anterior eccentricity, posterior eccentricity can provide information about deviations in corneal geometry. Its inclusion allows the model to consider shape information from the posterior surface in addition to posterior curvature and elevation.

The combination of EC_P with the other six maps creates a richer representation of corneal morphology for the multimodal learning framework.

(g) EC_P – Posterior eccentricity



The KeraSense framework considers multiple corneal representations to capture complementary information about corneal shape, elevation, thickness, and eccentricity. Figure 3.1 presents the seven corneal maps considered in the analysis: anterior sagittal curvature, posterior sagittal curvature, anterior elevation, posterior elevation, corneal thickness, anterior eccentricity, and posterior eccentricity. Each map represents a different aspect of corneal morphology and therefore contributes distinct information for the classification of normal, suspect, and keratoconic cases.

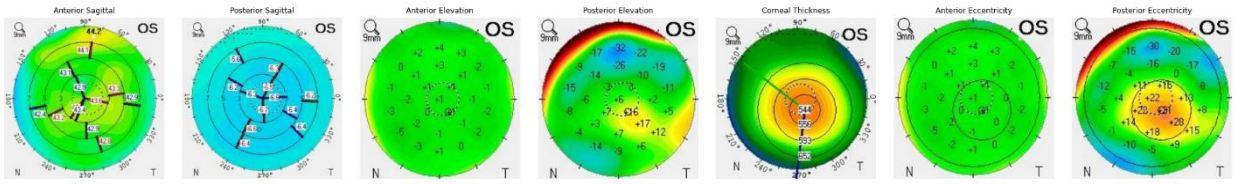


Figure 3.1: Example of the seven corneal maps used in the KeraSense framework.

3.4 Convolutional Neural Networks

A Convolutional Neural Network (CNN) is a deep learning architecture designed primarily for processing structured grid-like data such as images. CNNs are particularly effective for medical image analysis because they can learn spatial patterns without requiring every feature to be manually defined.

A typical CNN contains convolutional layers, activation functions, pooling operations, and fully connected or classification layers. The convolutional layers apply learnable filters to the input image and generate feature maps. During the early stages of the network, these features may represent simple patterns such as edges and local intensity changes. Deeper layers can combine these basic patterns into more complex structures.

The convolution operation can be represented conceptually as:

$$Y(i,j) = \sum_{m}\sum_{n} X\,(i+m, j+n)K(m,n)$$

where Xrepresents the input feature map, Krepresents the learnable convolution kernel, and Y represents the resulting feature map.

A nonlinear activation function is generally applied after convolution. The Rectified Linear Unit (ReLU) is commonly expressed as:

$$f(x) = \max(0, x)$$

Pooling operations can subsequently reduce the spatial dimensions of feature maps while retaining important information. This helps decrease computational requirements and allows deeper layers to operate on increasingly abstract representations.

For KeraSense, CNN-based feature extraction is used to learn visual characteristics from the corneal maps rather than depending entirely on manually designed features.

Figure 3.3: Basic concept of hierarchical feature extraction in a convolutional neural network.



3.5 Transfer Learning

Training a deep neural network from the beginning generally requires a large amount of labeled data and substantial computational resources. Medical imaging datasets are often considerably smaller than the datasets used to train large general-purpose vision models. Transfer learning provides a practical alternative.

In transfer learning, a model initially trained on a large source dataset is adapted to a different target problem. The pretrained network already contains useful low-level visual representations, which can then be refined using the target dataset.

A typical transfer-learning process involves replacing or adapting the original classification layer and subsequently fine-tuning some or all of the network parameters. The degree of fine-tuning can be adjusted depending on dataset size and similarity between the source and target domains.

For KeraSense, transfer learning provides a practical starting point for adapting a pretrained image-classification architecture to corneal image analysis. This approach reduces the need to learn all visual representations from randomly initialized parameters.

However, natural-image features and medical-image features are not identical. Consequently, transfer learning does not eliminate the need for appropriate preprocessing, training, validation, and evaluation.

3.6 EfficientNet-B0

EfficientNet is a family of convolutional neural networks designed to achieve a favorable balance between model accuracy and computational efficiency. Instead of increasing only the depth or width of a network, EfficientNet uses a compound scaling strategy that considers network depth, width, and input resolution together.

EfficientNet-B0 is the baseline model in the EfficientNet family. It provides a relatively compact architecture while retaining the ability to learn hierarchical image representations. This makes it suitable for applications where computational resources and dataset size need to be considered carefully.

In KeraSense, EfficientNet-B0 is used as the main feature-extraction backbone. The network receives the prepared corneal image representation and produces a learned feature representation that is subsequently used for classification.

The choice of EfficientNet-B0 is motivated by three practical considerations:

1. **Computational efficiency** compared with substantially larger architectures.
2. **Compatibility with transfer learning**, allowing pretrained weights to be utilized.
3. **Adequate feature-learning capability** for image-classification tasks.

The architecture should therefore be viewed as the feature-learning component of KeraSense rather than as the complete system.

3.7 Multimodal Feature Fusion

The seven corneal maps used in KeraSense contain different but related information. Multimodal feature fusion refers to the process of combining representations obtained from different input sources so that the final model can use complementary information.

In a feature-level fusion strategy, each input representation can first be processed to obtain a feature vector. These feature representations can then be combined into a unified representation:

$$F_{combined} = [F_1, F_2, \dots, F_n]$$

where $F_i$represents the learned feature vector associated with the i^{th}input representation.

For the KeraSense framework, the seven maps can be considered different views of the same underlying corneal structure. Their combination allows the model to learn relationships between curvature, elevation, thickness, and eccentricity characteristics.

The advantage of feature fusion is that the final representation does not depend exclusively on one type of corneal measurement. At the same time, fusion increases the dimensionality of the learned representation and can increase computational requirements. Therefore, the fusion strategy needs to be evaluated experimentally rather than assuming that additional inputs automatically produce better classification.

3.8 Multiclass Classification

The classification task in KeraSense involves three target classes:

- **Normal**
- **Suspect**
- **Keratoconus**

This is therefore a multiclass classification problem rather than a binary classification problem.
For a model with C classes, the output layer can produce a set of class scores:

$$z = [z_1, z_2, \dots, z_C]$$

These scores can be converted into class probabilities using the Softmax function:

$$P(y = i|x) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}$$

The predicted class is generally the class with the highest probability.
For three-class KeraSense classification:

$$C = 3$$

and the output corresponds to the three target categories.
This classification formulation is useful because the Suspect category represents an intermediate state rather than forcing all cases into either healthy or clearly diseased groups. It consequently provides a more challenging classification setting for evaluating the ability of the model to distinguish subtle structural differences.
Important: If your implementation actually performed a separate **binary experiment**, you can discuss binary classification as a secondary concept, but the main theoretical section should match the actual final model.

3.9 Grad-CAM and Explainable AI

Deep learning models can achieve strong classification performance while remaining difficult to interpret. Explainable AI provides techniques for examining the information that contributes to model predictions.
Gradient-weighted Class Activation Mapping, commonly known as **Grad-CAM**, is a visualization technique that identifies image regions associated with a particular model prediction. It uses gradients flowing into the final convolutional feature maps to estimate the importance of different spatial regions.
Conceptually, Grad-CAM calculates importance weights for feature maps using the gradients of the target class score. The resulting activation maps can be combined to produce a coarse localization map:

$$L^{c}_{\text{Grad-CAM}} = ReLU\left(\sum_{k} \alpha_k^c \, A^k\right)$$

where A^krepresents the k^{th}feature map and α_k^crepresents its importance for class c
The ReLU operation retains the regions that have a positive influence on the selected prediction. The resulting heatmap can then be resized and overlaid on the original corneal image.
For KeraSense, Grad-CAM is used to examine whether the model focuses on visually meaningful regions when making its prediction. For example, a heatmap may reveal whether the model concentrates around regions displaying unusual curvature, elevation, or other structural characteristics.
Grad-CAM does not establish that a highlighted region is clinically causal. Instead, it provides an additional method for investigating model behavior and identifying potentially meaningful patterns.

3.10 Performance Evaluation Metrics

The performance of the KeraSense classification model should be evaluated using multiple metrics rather than accuracy alone. This is particularly important for medical classification

because class distributions may not always be perfectly balanced and different types of classification errors can have different implications.

Accuracy

Accuracy represents the proportion of correctly classified samples among all evaluated samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

For multiclass classification, accuracy is calculated as the total number of correct predictions divided by the total number of predictions.

Precision

Precision measures the proportion of predicted positive cases that are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

For multiclass classification, precision can be calculated independently for each class and then averaged using an appropriate averaging strategy.

Recall

Recall, also referred to as sensitivity, measures the proportion of actual positive cases that are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

Recall is particularly important when missing a relevant abnormal case is considered an important error.

F1-Score

The F1-score combines precision and recall through their harmonic mean:

$$F1 = 2\left(\frac{Precision \times Recall}{Precision + Recall}\right)$$

For a multiclass problem, macro-F1 can be useful because it gives equal importance to each class.

Confusion Matrix

A confusion matrix provides a class-by-class representation of model predictions. For KeraSense, the matrix can be organized as:

Actual / Predicted	Normal	Suspect	Keratoconus
Normal	Correct	Error	Error
Suspect	Error	Correct	Error
Keratoconus	Error	Error	Correct

This representation is particularly useful for identifying whether the model has difficulty distinguishing **Suspect** cases from Normal or Keratoconus cases.

Macro-Averaged Metrics

For three-class classification, macro-averaged precision, recall, and F1-score can provide a balanced view of performance because the metric is calculated separately for each class and then averaged.

For example:

$$F1_{macro} = \frac{F1_{Normal} + F1_{Suspect} + F1_{KC}}{3}$$

This prevents a highly represented class from completely dominating the overall metric

CHAPTER 4

KERASENSE SYSTEM DESIGN AND METHODOLOGY

4.1 Overall System Workflow

The KeraSense system was designed as a multimodal image analysis framework for identifying keratoconus from multiple representations of the cornea. Instead of relying on a single corneal map, the system processes seven complementary maps belonging to the same eye. These maps describe different properties of corneal curvature, elevation, thickness, and eccentricity. Using the maps together allows the model to consider structural information that may not be sufficiently represented in an individual map.

The complete workflow begins with the organization and verification of the dataset at the eye level. Each eye is represented by seven corresponding images: anterior sagittal curvature (Sag_A), posterior sagittal curvature (Sag_P), anterior elevation (Elv_A), posterior elevation (Elv_P), corneal thickness (CT_A), anterior eccentricity (EC_A), and posterior eccentricity (EC_P). The seven images belonging to one eye are treated as a single sample throughout the data preparation and splitting stages.

After data verification, the images undergo a common preprocessing procedure. The images are converted into a suitable RGB representation, resized to 224 times 224 pixels, transformed into tensors, and normalized using ImageNet normalization values. The processed images are then passed through a shared EfficientNet-B0 feature extractor. Each map produces a 1280-dimensional feature representation from the final feature extraction stage.

The seven feature vectors are concatenated to form a single 8960-dimensional representation of the eye. This fused representation is supplied to a multilayer perceptron (MLP), which learns the relationship between the combined corneal features and the target classes. The final classification layer produces probabilities for the two output categories, **KC** and **Non-KC**.

Grad-CAM is subsequently applied to the individual map branches to visualize the image regions that contribute strongly to the model's prediction. This provides an interpretable view of the learned decision process and helps examine whether the network is responding to meaningful corneal patterns.

The overall processing sequence can therefore be summarized as:

Seven corneal maps → Image preprocessing → Shared EfficientNet-B0 → Seven 1280-dimensional feature vectors → Feature concatenation → 8960-dimensional fused representation → MLP classifier → Softmax → KC/Non-KC prediction → Grad-CAM interpretation

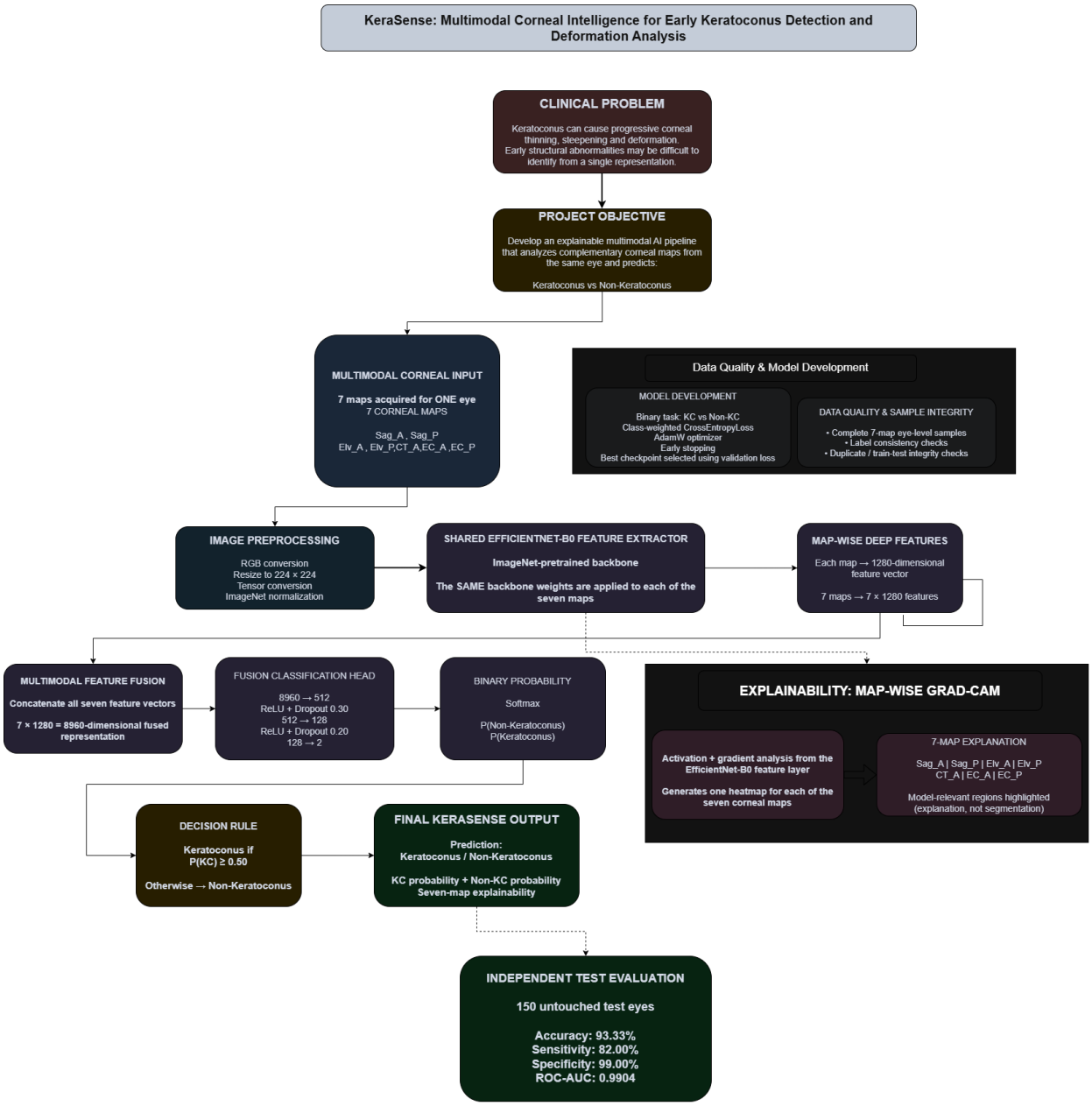


Figure 4.1 – Overall KeraSense Architecture

4.2 Dataset Description

The KeraSense model was developed using a publicly available keratoconus image dataset containing multiple corneal maps for each eye. The dataset is organized into three original clinical categories: **Normal, Suspect, and Keratoconus**. Each eye is associated with seven corneal map images, allowing the relationship between different representations of the same cornea to be retained during model development.

The complete dataset contains **4,011 image files**. The training/validation portion contains **2,961 images**, while the independent test portion contains **1,050 images**. Since seven maps are available for each eye, these image counts correspond to **423 eyes** in the training/validation portion and **150 eyes** in the independent test portion.

The training/validation images consist of 1,050 Normal images, 861 Suspect images, and 1,050 Keratoconus images. The independent test set contains an equal number of images from each

original class, with 350 images corresponding to Normal, 350 to Suspect, and 350 to Keratoconus cases.

Table 4.1 – Dataset Distribution

Dataset portion	Normal	Suspect	Keratoconus	Total images	Approx. eyes
Training/Validation	1050	861	1050	2961	423
Independent Test	350	350	350	1050	150
Total	**1400**	**1211**	**1400**	**4011**	**573**

The original three-class organization is retained during dataset verification because it reflects the source dataset. For the final KeraSense classification task, these categories are mapped into two clinically meaningful groups, as described in Section 4.6.

4.3 Dataset Organization and Seven-Map Eye Structure

A key characteristic of the dataset is that the seven images associated with an eye are not treated as independent observations. They collectively describe the same corneal structure and therefore need to remain together during dataset preparation.

For every eye, the following seven maps are considered:

1. **Sag_A** – Anterior Sagittal Curvature
2. **Sag_P** – Posterior Sagittal Curvature
3. **Elv_A** – Anterior Elevation
4. **Elv_P** – Posterior Elevation
5. **CT_A** – Corneal Thickness
6. **EC_A** – Anterior Eccentricity
7. **EC_P** – Posterior Eccentricity

The seven-map organization is important because separating maps from the same eye during training and validation could cause information from one eye to appear in both sets. Such an arrangement would make the evaluation overly optimistic. KeraSense therefore uses the **eye as the basic unit of data organization and splitting**.

For the training/validation portion, the 423 available eyes were divided into **338 training eyes and 85 validation eyes**. The independent test set contains **150 additional eyes** that are kept separate from model development.

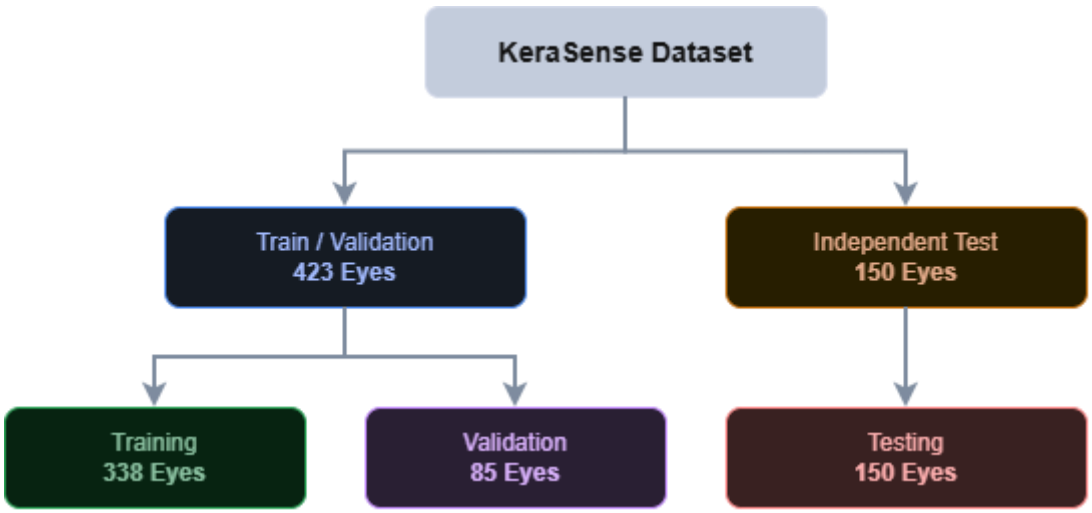


Figure 4.2 – Dataset Organization

4.4 Data Quality and Integrity Verification

Before model training, the dataset was examined to ensure that the image organization was consistent with the intended seven-map structure. This step was necessary because a multimodal model depends on correct correspondence between the different maps belonging to the same eye.

The verification process checked whether the required map types were available for each eye and whether the corresponding image files could be associated with a common eye identifier. The number of images was also compared with the expected number obtained from the seven-map structure.

For the training/validation portion, 2,961 images correspond to 423 eyes because:

$$423 \times 7 = 2961$$

Similarly, the independent test set contains:

$$150 \times 7 = 1050$$

These relationships provide a useful consistency check on the dataset organization.

An additional **data leakage check** was performed at the eye level. The identifiers belonging to the training, validation, and independent test groups were compared to ensure that the same eye did not occur in more than one partition. This is particularly important for corneal imaging because seven maps from the same eye can contain highly related visual patterns.

The integrity checks confirmed that the seven-map grouping was maintained and that the independent test eyes were kept separate from the training and validation groups.

4.5 Image Preprocessing

The images supplied to the neural network must have a consistent format and spatial resolution. A common preprocessing procedure was therefore applied to the seven map types before feature extraction.

First, the input images were represented in RGB format so that they could be processed consistently by the pretrained EfficientNet-B0 network. Each image was then resized to (224 times 224) pixels. This resolution corresponds to the input size used by the EfficientNet-B0 configuration adopted in the KeraSense implementation.

After resizing, the images were converted into tensors. Pixel values were then normalized using the mean and standard deviation associated with ImageNet-pretrained models. For a normalized pixel value, the transformation can be expressed as:

$$x' = \frac{x - \mu}{\sigma}$$

where x represents the input pixel value, μ represents the channel-wise mean, and σ represents the corresponding standard deviation.

The same preprocessing procedure was applied across all seven map types. Maintaining a consistent transformation prevents differences introduced by preprocessing from becoming an unintended source of variation between the input branches.

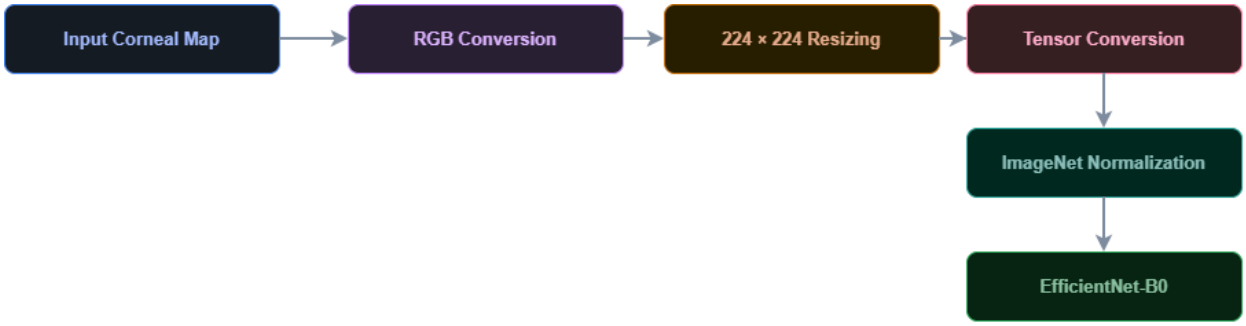


Figure 4.4 – Image Preprocessing Pipeline

4.6 Eye-Level Data Splitting and Binary Labeling

4.6.1 Eye-Level Data Splitting

The data were divided at the **eye level rather than the individual image level**. This decision is important because each eye contributes seven related images. Randomly distributing individual maps between training and validation sets could allow different representations of the same eye to appear in both groups.

The 423 eyes available in the training/validation portion were divided into **338 training eyes and 85 validation eyes**. The 150 eyes belonging to the independent test set were not used during training or validation.

The resulting organization is:

Partition	Number of eyes	Role
Training	338	Model parameter learning
Validation	85	Model monitoring and selection
Independent Test	150	Final performance evaluation
Total	**573**	—

This arrangement ensures that the final test evaluation is performed on eyes that were not involved in model optimization.

4.6.2 Binary Labeling

Although the source dataset contains three categories—Normal, Suspect, and Keratoconus—the final KeraSense experiment uses a binary classification formulation.

The **Keratoconus** category is assigned to the **KC** class. The Normal and Suspect categories are grouped into the **Non-KC** class.

The mapping can be expressed as:

$$y = \begin{cases} 0, & \text{Normal or Suspect} \rightarrow \text{Non-KC} \\ 1, & \text{Keratoconus} \rightarrow \text{KC} \end{cases}$$

This formulation allows the model to focus on the primary screening question: whether the available corneal maps contain sufficient information to distinguish a keratoconic eye from an eye that is not categorized as keratoconus.

The binary grouping also preserves the Suspect cases instead of removing them from the dataset. This is useful because suspect cases represent a visually and clinically relevant intermediate group, even though they are grouped with Non-KC for the final binary experiment.

4.7 KeraSense Model Architecture

The central component of KeraSense is a **multi-map feature extraction and fusion architecture**. The model uses EfficientNet-B0 as the feature extractor for each of the seven corneal maps.

Rather than building seven completely independent deep networks, KeraSense uses a **shared EfficientNet-B0 feature extractor**. Each of the seven maps is passed through the same feature extraction architecture, allowing the model to learn a common visual representation while still processing each corneal map separately.

The architecture can be represented as:

$$I_i \rightarrow E(I_i) \rightarrow F_i$$

where \(I_i\) is the \(i^{th}\) corneal map, \(E\) represents the shared EfficientNet-B0 feature extractor, and \(F_i\) is the resulting feature vector.

For the seven maps:

$$F_{Sag_A}, F_{Sag_P}, F_{Elv_A}, F_{Elv_P}, F_{CT_A}, F_{EC_A}, F_{EC_P}$$

are generated separately.

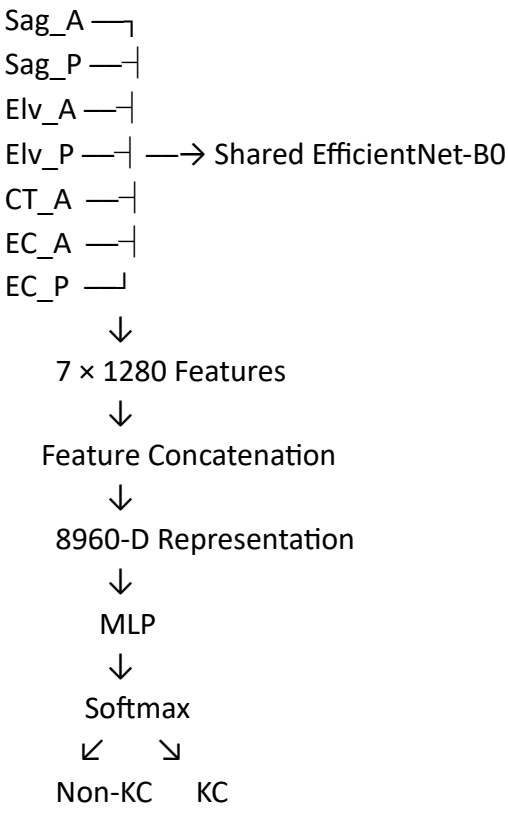
Each branch produces a **1280-dimensional feature vector**. Therefore, the complete feature representation contains:

$$7 \times 1280 = 8960$$

features.

The fused representation is then provided to the MLP classification head. The final layer produces two output values corresponding to the KC and Non-KC classes.

The overall architecture can be summarized as:



4.8 Multimodal Feature Extraction and Fusion

The main methodological component of KeraSense is the combination of information obtained from multiple corneal maps.

Each map captures a different property of the cornea. For example, curvature maps describe the shape of the anterior or posterior surface, elevation maps describe deviations relative to a reference surface, thickness maps provide information about the corneal thickness distribution, and eccentricity maps represent characteristics associated with the shape of the corneal surface. Processing these maps separately before fusion allows the network to learn map-specific visual features. The resulting feature vectors are then combined at a higher representation level.

Let the feature vectors obtained from the seven maps be:

$$F_1, F_2, F_3, F_4, F_5, F_6, F_7$$

where each vector has a dimensionality of 1280. Feature fusion is performed through concatenation:

$$F_{fused} = [F_1; F_2; F_3; F_4; F_5; F_6; F_7]$$

Therefore:

$$dim(F_{fused}) = 7 \times 1280 = 8960$$

The resulting 8960-dimensional vector represents the combined information available from all seven corneal maps. This fused representation is passed to the MLP classifier, which learns relationships between the combined features and the final KC/Non-KC decision.

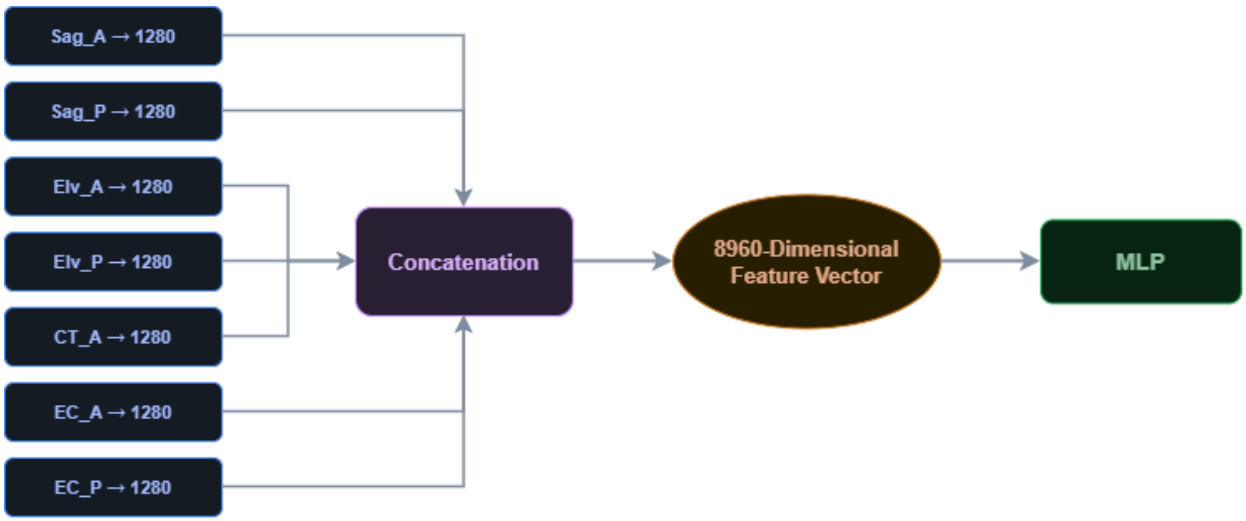


Figure 4.5 – Feature Fusion Process

4.9 Model Training Strategy

The final KeraSense model was trained using the binary classification setup described in the previous sections. The seven corneal maps corresponding to each eye were processed together as a single training sample. A batch size of 8 eyes was used, meaning that each training batch contained 8 complete eye samples, with seven maps processed for each eye. mip

The model was optimized using the AdamW optimizer with an initial learning rate of $1 \times 10^{-4}$and a weight decay of 1×10^{-4}. All parameters of the EfficientNet-B0 backbone were

allowed to participate in training, rather than keeping the pretrained feature extractor completely frozen. mip

Because the binary training data were not perfectly balanced, a class-weighted Cross-Entropy Loss was used. The training set contained 218 Non-Keratoconus eyes and 120 Keratoconus eyes. Inverse-frequency weighting was applied to give greater importance to the minority Keratoconus class. The resulting normalized class weights were approximately 0.71 for Non-Keratoconus and 1.29 for Keratoconus. mip

The binary model was configured to train for a maximum of 20 epochs. To reduce unnecessary training after the validation performance stopped improving, early stopping with a patience of 5 epochs was employed. At each epoch, the training loss and accuracy were calculated, followed by evaluation on the validation set. The model state corresponding to the lowest validation loss was saved as the best checkpoint. mip

The training configuration used for the final binary KeraSense experiment is summarized in Table 4.2.

Table 4.2 – Training Configuration of the KeraSense Model

Parameter	Configuration
Backbone	EfficientNet-B0
Pretrained weights	ImageNet
Input maps per eye	7
Input resolution	\(224 \times 224\)
Feature dimension per map	1280
Fused feature dimension	8960
Classification task	Binary
Output classes	Non-Keratoconus, Keratoconus
Loss function	Class-weighted Cross-Entropy Loss
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Batch size	8 eyes
Maximum epochs	20
Early stopping patience	5 epochs
Best-model criterion	Minimum validation loss
Backbone training	All layers trainable

4.10 Grad-CAM Explainability

Deep learning models can achieve strong classification performance while providing limited information about the visual evidence used to reach a prediction. KeraSense therefore incorporates **Gradient-weighted Class Activation Mapping (Grad-CAM)** to examine the spatial regions that contribute to a particular prediction.

Grad-CAM uses the gradients of the selected class with respect to feature maps in a convolutional layer. The importance of each feature map is estimated by globally averaging its gradients. The resulting weights are then combined with the corresponding activation maps.

The Grad-CAM representation can be expressed as:

$$L^{c}_{\text{Grad-CAM}} = ReLU\left(\sum_{k} \alpha_k^c \, A^k\right)$$

where A^krepresents the k^{th}activation map and α_k^crepresents its importance for class c.
The ReLU operation retains the regions that provide positive evidence for the selected prediction.
In KeraSense, Grad-CAM is particularly useful because the model processes seven different corneal representations. Heatmaps can therefore be examined for individual map types to determine which regions of the corneal image contribute to the model's decision.
For example, Grad-CAM visualizations can be generated for the anterior elevation, posterior elevation, thickness, or curvature maps and compared with the corresponding original images. This does not prove that the highlighted region represents a clinically diagnostic feature, but it provides an additional method for inspecting the behavior of the trained network.
The explainability stage is therefore used as a **model interpretation tool rather than as an independent diagnostic method**.

4.11 Inference and Decision Process

After training, an unseen eye is processed using the same sequence applied during model development. The seven maps associated with the eye are first identified and subjected to the same preprocessing operations.
The processed maps are then passed individually through the shared EfficientNet-B0 feature extractor. Each map produces a 1280-dimensional representation. These seven representations are concatenated to create the 8960-dimensional eye-level feature vector.
The MLP classifier processes this vector and produces two output probabilities:

- **Non-KC probability**
- **KC probability**

The final prediction is determined by selecting the class with the higher probability.
The inference procedure can be represented as:

Unseen Eye
↓
Identify 7 Corneal Maps
↓
Apply Common Preprocessing
↓
Shared EfficientNet-B0
↓
7 × 1280 Feature Vectors
↓
Concatenate Features
↓
8960-Dimensional Representation
↓
MLP Classifier
↓
Softmax Probabilities
↓
KC / Non-KC
↓
Grad-CAM Interpretation

The same eye-level structure is maintained during inference so that all seven maps contribute to one final prediction.

4.12 Evaluation Methodology

The final performance of KeraSense is evaluated using the independent test set containing **150 previously unseen eyes**. Because these eyes are kept separate from training and validation, the test results provide an estimate of the model's performance on unseen cases.

Several classification metrics are considered to provide a more complete evaluation than accuracy alone.

4.12.1 Accuracy

Accuracy represents the proportion of correctly classified eyes among all evaluated eyes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

4.12.2 Precision

Precision measures how many of the eyes predicted as KC actually belong to the KC class.

$$Precision = \frac{TP}{TP + FP}$$

A higher precision indicates fewer false-positive KC predictions.

4.12.3 Recall

Recall measures the proportion of actual KC eyes that are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

Recall is particularly important in a screening-oriented setting because missed keratoconus cases correspond to false negatives.

4.12.4 F1-Score

The F1-score combines precision and recall using their harmonic mean:

$$F1 = 2\left(\frac{Precision \times Recall}{Precision + Recall}\right)$$

This metric provides a balanced measure when both false-positive and false-negative errors need to be considered.

4.12.5 Confusion Matrix

A confusion matrix is used to examine the distribution of correct and incorrect predictions across the two classes. For the binary KeraSense experiment, the matrix contains:

- True Non-KC
- False KC
- False Non-KC
- True KC

The confusion matrix provides more detailed information than a single accuracy value because it shows the types of classification errors made by the model.

4.12.6 ROC-AUC

Where probability outputs are retained, the Receiver Operating Characteristic (ROC) curve can be used to examine the relationship between the true-positive rate and false-positive rate at different classification thresholds.

The Area Under the ROC Curve (AUC) summarizes the model's ability to distinguish KC from Non-KC across those thresholds. A value closer to 1 indicates stronger class separation, whereas a value near 0.5 indicates performance close to random discrimination.

4.13 Chapter Summary

This chapter presented the complete methodology used to develop the KeraSense system. The approach begins by treating the **eye rather than the individual image as the fundamental data unit**, thereby preserving the relationship between the seven corneal maps associated with each eye. Dataset integrity was verified before preprocessing, and the data were divided into training, validation, and independent test groups without allowing the same eye to appear across different partitions.

The proposed architecture uses a **shared EfficientNet-B0 feature extractor** to process the seven corneal maps independently. Each map produces a 1280-dimensional representation, resulting in an **8960-dimensional fused feature vector** after concatenation. An MLP then performs the final binary classification between KC and Non-KC.

The methodology also incorporates Grad-CAM to provide visual insight into the regions influencing model predictions. Finally, accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC are used to evaluate the model on previously unseen test eyes.

Overall, the methodology combines **multimap corneal representation, shared deep feature extraction, eye-level data handling, feature-level fusion, binary classification, and explainability** into a single KeraSense pipeline.

CHAPTER 5: IMPLEMENTATION

This chapter describes the practical implementation of the KeraSense system, including the computational environment, dataset loading, preprocessing pipeline, model construction, training procedure, threshold analysis, independent testing, Grad-CAM generation, and the final user interface. The implementation was carried out in Python using PyTorch and supporting image-processing and machine-learning libraries.

5.1 Software and Hardware Environment

The KeraSense implementation was developed in a GPU-enabled Python environment. **PyTorch** was used as the primary deep learning framework, while **Torchvision** provided the pretrained EfficientNet-B0 architecture and image transformation utilities. **Pandas** and **NumPy** were used for dataset organization and numerical operations, while **Pillow** was used for image loading. Scikit-learn was used for dataset splitting and performance evaluation.

The model was trained using an **NVIDIA Tesla T4 GPU** with CUDA acceleration. GPU execution was used because the KeraSense architecture processes seven images for every eye and performs feature extraction through the EfficientNet-B0 backbone for each map.

Table 5.1 – Software and Hardware Environment

Component	Configuration
Programming language	Python 3.13.15
Deep learning framework	PyTorch 2.11.0
Data handling	Pandas

Machine learning utilities	Scikit-learn 1.9.1
Visualization	Matplotlib
Interactive interface	IPyWidgets
Model architecture	EfficientNet-B0
CUDA	CUDA 13.0 (cu130)
GPU	NVIDIA Tesla T4
Computer vision/model library	Torchvision
Image handling	Pillow
Numerical processing	NumPy

The GPU environment was verified before model development by checking CUDA availability and identifying the available GPU device. The system reported CUDA availability as True and detected the Tesla T4 accelerator.

Figure 5.1 – Development Environment and GPU Configuration.

```
Using Colab cache for faster access to the 'keratoconus-detection' dataset.
KeraSense environment ready.
Device: cuda
GPU: Tesla T4
Dataset path: /kaggle/input/keratoconus-detection
```

5.2 Dataset Preparation and Loading

The dataset preparation stage converts the original image collection into a structured representation that can be directly consumed by the KeraSense model. Since each eye is represented by seven different corneal maps, the implementation does not treat individual image files as independent samples.

The dataset-building procedure first traverses the relevant directories and identifies images corresponding to the seven predefined map types:

- Sag_A
- Sag_P
- Elv_A
- Elv_P
- CT_A
- EC_A
- EC_P

For every image, the map identifier is extracted from its filename. The remaining portion of the filename is used as the eye/sample identifier. Images sharing the same sample identifier are grouped together into a single record.

A sample is considered complete only when all seven required maps are available. Incomplete records are excluded from the model dataset. This ensures that every sample supplied to KeraSense contains the complete seven-map representation required by the architecture. mip

The original class labels are initially represented as:

$$Normal = 0, \qquad Suspect = 1, \qquad Keratoconus = 2$$

The final binary model subsequently maps Normal and Suspect to Non-Keratoconus and retains Keratoconus as the positive class.

The training/validation dataset was divided using a **stratified 80:20 split with random seed 42**. This produced 338 training eyes and 85 validation eyes, while the 150-eye independent test set remained separate.

5.3 Preprocessing Implementation

The preprocessing pipeline was implemented using the torchvision.transforms module. Each corneal map is loaded using Pillow and converted to RGB before being transformed into the format expected by EfficientNet-B0.

The implemented preprocessing sequence consists of four main operations:

1. **RGB conversion**
2. **Resizing to** 224×224 **pixels**
3. **Conversion to a PyTorch tensor**
4. **ImageNet normalization**

The normalization parameters used in the implementation are:

$$\mu = [0.485, 0.456, 0.406]$$

and

$$\sigma = [0.229, 0.224, 0.225]$$

For each eye, the seven processed images are stacked into a tensor with the shape:

$$(7, 3, 224, 224)$$

where the first dimension represents the seven corneal maps, the second represents the RGB channels, and the final two dimensions represent image height and width. mip

The same preprocessing operation is applied to all seven map types so that differences between maps are determined by their actual corneal information rather than inconsistent input processing.

5.4 KeraSense Model Implementation

The KeraSense model was implemented as a custom PyTorch neural network using a pretrained **EfficientNet-B0** backbone.

The EfficientNet-B0 model was initialized with the pretrained ImageNet weights available through Torchvision. Its original ImageNet classification layer was removed and replaced with an identity operation. This allows the network to provide its learned feature representation instead of directly producing ImageNet class predictions.

Each of the seven corneal maps is passed through the same EfficientNet-B0 instance. Consequently, the backbone parameters are shared across all seven map branches.

For each map, EfficientNet-B0 produces a **1280-dimensional feature vector**. The seven vectors are concatenated:

$$7 \times 1280 = 8960$$

The resulting 8960-dimensional vector is processed by the KeraSense classification head:

$$8960 \rightarrow 512 \rightarrow 128 \rightarrow 2$$

The implemented head contains:

- Linear layer: 8960 → 512
- ReLU activation
- Dropout: 0.30
- Linear layer: 512 → 128
- ReLU activation
- Dropout: 0.20
- Linear layer: 128 → 2

The final two outputs correspond to:

- **Non-Keratoconus**

- **Keratoconus**

Softmax is applied during probability estimation to convert the two output logits into class probabilities. mip

Implementation Flow

7 Corneal Maps
↓
Shared EfficientNet-B0
↓
7 × 1280 Features
↓
Concatenation
↓
8960 Features
↓
Linear 8960 → 512
↓
ReLU + Dropout(0.30)
↓
Linear 512 → 128
↓
ReLU + Dropout(0.20)
↓
Linear 128 → 2
↓
Non-KC / KC

This implementation preserves the relationship between the seven maps while allowing the network to learn a joint representation for the eye.

5.5 Training and Validation Implementation

The final binary KeraSense model was trained using PyTorch DataLoader objects with a **batch size of 8 eyes**. Training batches were shuffled, while validation batches were processed without shuffling. Two worker processes and pinned memory were used for the data loaders. mip

The model was optimized using **AdamW** with:

$$Learning\ Rate = 1 \times 10^{-4}$$

and

$$Weight\ Decay = 1 \times 10^{-4}$$

A **class-weighted Cross-Entropy Loss** was used to compensate for the imbalance between the Non-Keratoconus and Keratoconus classes in the binary training set. The training data contained 218 Non-Keratoconus eyes and 120 Keratoconus eyes. Inverse-frequency weighting was applied before normalizing the weights. mip

The maximum training duration was **20 epochs**, with **early stopping patience of 5 epochs**. After every epoch, the model was evaluated on the validation set. When a lower validation loss was obtained, the model parameters were saved as the current best checkpoint. Training was terminated when the validation loss failed to improve for five consecutive epochs. mip

Importantly, **no learning-rate scheduler was used in the final binary training implementation**. The learning rate remained fixed at 1×10^{-4} The best saved model was subsequently restored before independent test evaluation.

Table 5.2 – Final Binary Training Implementation

Parameter	Value
Training samples	338 eyes
Validation samples	85 eyes
Batch size	8
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Loss	Weighted Cross-Entropy
Maximum epochs	20
Early stopping patience	5
Learning-rate scheduler	None
Model selection criterion	Minimum validation loss
Checkpoint	Best validation-loss model

The final training code restored the best checkpoint before proceeding to the independent test stage.

5.6 Validation-Based Threshold Selection

The model produces a probability representing the likelihood of the Keratoconus class. Rather than directly treating the raw output as a class label, a decision threshold is applied to this probability.

The validation set was used to examine the behavior of different KC probability thresholds. Thresholds from **0.10 to 0.90 in increments of 0.05** were evaluated. For every threshold, validation accuracy, sensitivity, and specificity were calculated. mip

The threshold analysis was performed before the final independent test evaluation so that the test set would not be used for threshold tuning.

For the final KeraSense inference and independent binary evaluation, the decision threshold was **locked at 0.50**:

$$\hat{y} = \begin{cases} KC, & P(KC) \geq 0.50 \\ Non\text{-}KC, & P(KC) < 0.50 \end{cases}$$

Once fixed, the threshold was not changed during the final test evaluation. The same value is also used by the final inference function and the user interface.

This separation between threshold analysis and final testing prevents the independent test set from influencing the operating decision rule.

5.7 Independent Test Evaluation

After the best binary model was restored, the independent test dataset was loaded using the same preprocessing procedure. The test loader contained the previously untouched test eyes and used a batch size of 8 with no shuffling. mip

For each test eye, the model generated two output logits. Softmax was then applied to obtain the probability of the Keratoconus class:

$$P(KC) = Softmax(output)_1$$

The locked threshold of 0.50 was then applied to produce the final binary prediction.

The implementation calculates the following measures:

- Accuracy
- Sensitivity

- Specificity
- Positive Predictive Value (PPV)
- Negative Predictive Value (NPV)
- ROC-AUC
- Confusion matrix
- Detailed classification report

The confusion matrix is arranged as:

$$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$$

where the positive class represents Keratoconus.

ROC-AUC is calculated using the continuous KC probabilities rather than thresholded predictions, allowing the model's ranking ability to be evaluated independently of the selected decision threshold. mip

Important: the actual numerical test results should be presented in **Chapter 6 – Results and Analysis**, rather than repeated here. Chapter 5 explains the implementation of the evaluation procedure.

5.8 Grad-CAM Implementation

Grad-CAM was implemented to provide a visual interpretation of the features used by the trained KeraSense model.

For a selected test eye, the seven corneal maps are first passed through the binary KeraSense model. During the forward pass, activations from the selected convolutional layer of the EfficientNet-B0 backbone are recorded. The gradient of the KC probability with respect to these activations is then calculated through backpropagation.

For each of the seven map branches, the gradients are globally averaged across their spatial dimensions to obtain feature importance weights. These weights are multiplied with the corresponding activation maps and summed to generate the Grad-CAM representation. A ReLU operation is applied to retain positive contributions, after which the heatmap is normalized to the range [0,1].

The generated heatmaps are resized to the original 224×224representation and overlaid on the corresponding corneal maps. Therefore, a single eye produces **seven individual Grad-CAM maps**, one for each input representation.

The implementation also produces a combined visualization containing the original seven maps and their corresponding Grad-CAM overlays. mip

This implementation allows the contribution of different regions to be examined separately for Sag_A, Sag_P, Elv_A, Elv_P, CT_A, EC_A, and EC_P rather than producing only one explanation for the complete eye.

5.9 KeraSense User Interface

A lightweight interactive interface was implemented using **IPyWidgets**. The interface provides seven individual upload controls corresponding to the seven required corneal maps:

- Sag_A
- Sag_P
- Elv_A
- Elv_P
- CT_A
- EC_A

- EC_P

The user is required to provide one image for each map. The implementation checks whether every required input has been supplied before executing the prediction process. Uploaded files are temporarily stored and passed to the reusable kerasense_predict() inference function.

The prediction function verifies that exactly seven maps are provided and processes them in the required map order. The same resizing and ImageNet normalization used during model development are applied before inference.

After processing the seven maps, the interface displays:

- Predicted class
- Keratoconus probability
- Non-Keratoconus probability
- Decision threshold
- Number of maps analyzed

The final interface therefore provides a simple seven-input workflow while retaining the complete multimodal structure of the KeraSense model.

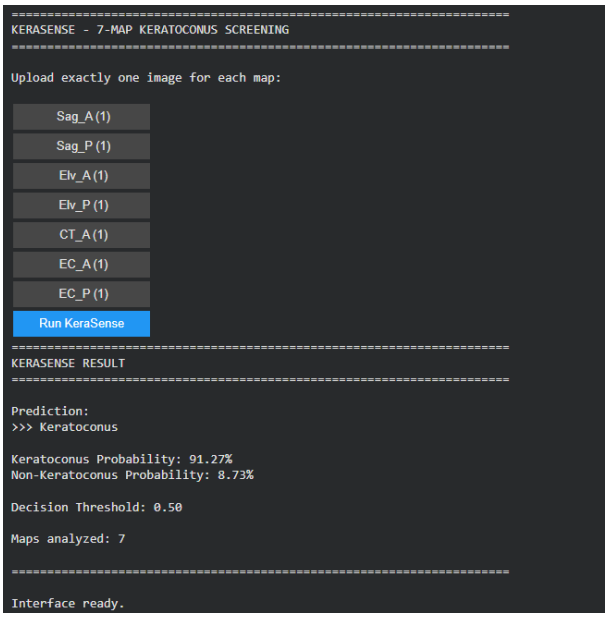


Figure 5.3: KeraSense seven-map inference interface implemented using IPyWidgets.

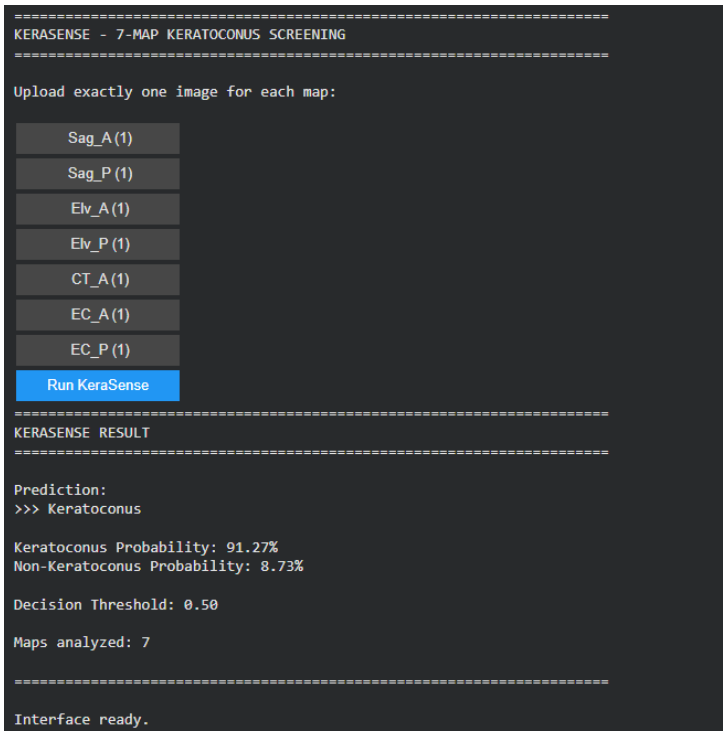


Figure 5.4: Example KeraSense inference output for a seven-map corneal input.

The displayed probabilities are generated from the Softmax output of the binary model, while the final class decision is determined using the locked KC probability threshold of 0.50.

5.10 Implementation Summary

The KeraSense implementation integrates dataset organization, seven-map image processing, deep feature extraction, feature fusion, binary classification, threshold-based decision making, independent evaluation, explainability, and interactive inference into a single workflow.

The implementation treats an **eye as one complete multimodal sample**, with its seven corneal maps processed using a shared EfficientNet-B0 backbone. The extracted 1280-dimensional representations are concatenated into an 8960-dimensional feature vector and passed through the classification head. Training uses AdamW, class-weighted Cross-Entropy Loss, a learning rate of 1×10^{-4}batch size 8, and early stopping based on validation loss.

The final model is evaluated on an independent set of previously unseen eyes using a locked KC probability threshold of 0.50. Grad-CAM provides map-level visual explanations, while the interactive interface allows all seven required corneal maps to be supplied together for a final KeraSense prediction.

This implementation establishes the practical foundation for the experimental results presented in the next chapter.

CHAPTER 6 : RESULTS AND ANALYSIS

This chapter presents the experimental results obtained from the final KeraSense binary classification model. The analysis covers the dataset distribution, training and validation behavior, threshold selection, independent test performance, confusion matrix, ROC-AUC analysis, Grad-CAM explanations, representative predictions, and comparison with earlier experiments.

The final evaluation was performed on an **independent set of 150 previously unseen eyes**. Each eye contained seven corresponding corneal maps. For the binary experiment, Normal and

Suspect cases were grouped as **Non-Keratoconus (Non-KC)**, while Keratoconus cases formed the **KC** class. The final model achieved **93.33% accuracy and an ROC-AUC of 0.9904** on the independent test set.

6.1 Dataset Statistics

The complete dataset contained 573 eyes, represented by 4,011 image files. The development portion consisted of 423 eyes and the independent test portion contained 150 eyes. Since each eye was represented by seven maps, the image counts correspond exactly to the number of eyes multiplied by seven.

The original three-class distribution was:

Table 6.1 – Dataset Distribution

Split	Normal	Suspect	Keratoconus	Total
Development	150	123	150	423
Independent Test	50	50	50	150
Total	**200**	**173**	**200**	**573**

For the final binary experiment, Normal and Suspect eyes were combined into the Non-KC category.

Split	Non-KC	KC	Total
Training	218	120	338
Validation	55	30	85
Independent Test	100	50	150
Total	**373**	**200**	**573**

The development set was divided into 338 training eyes and 85 validation eyes using a stratified split. The independent test set was kept completely separate from this process.

The binary training distribution was moderately imbalanced, with 218 Non-KC and 120 KC eyes in the training set. This imbalance was addressed during training through class-weighted Cross-Entropy Loss rather than by removing samples.

6.2 Training and Validation Performance

The final KeraSense model converged relatively quickly during training. The training and validation curves provide an indication of how the model learned from the development data and whether overfitting occurred.

The best checkpoint was obtained at **epoch 9**, where the validation loss reached **0.0525** and validation accuracy reached **98.82%**. The model was restored from this checkpoint before the independent test evaluation.



Figure 6.1: Training and validation accuracy curves and loss across epochs for the final KeraSense binary model. The best model was obtained at epoch 9 based on validation loss.

The training behavior shows that the model learned the classification task rapidly. Training accuracy approached a very high level during the later epochs, while validation performance reached its strongest point around epoch 9. After this point, the validation loss began to increase even though training performance continued to improve.

This difference between training and validation behavior indicates the beginning of overfitting. The use of early stopping prevented later model states from replacing the better-performing epoch-9 checkpoint.

The training results therefore support the decision to select the model according to **validation loss rather than training accuracy**.

6.3 Threshold Selection Results

The KeraSense model produces a probability for the Keratoconus class rather than directly producing a clinical decision. A threshold is therefore required to convert the probability into a binary prediction.

Thresholds ranging from **0.10 to 0.90** were evaluated on the validation set at intervals of 0.05. For each threshold, accuracy, sensitivity, and specificity were calculated.

The final decision threshold was fixed at:

$$\boxed{Threshold = 0.50}$$

The decision rule was:

$$Prediction = \begin{cases} KC, & P(KC) \geq 0.50 \\ Non\text{-}KC, & P(KC) < 0.50 \end{cases}$$

The threshold was selected using the validation data and then **locked before the independent test evaluation**. This prevents the test set from influencing the decision rule.

A lower threshold would generally make the classifier more sensitive to KC cases but could increase false-positive predictions. Conversely, a higher threshold would tend to reduce false positives while increasing the possibility of missing KC cases.

The selected value of 0.50 therefore provides a straightforward operating point and was retained consistently for the final test and user-interface predictions.

6.4 Independent Test Results

The final KeraSense model was evaluated on **150 independent test eyes**, consisting of 100 Non-KC and 50 KC eyes. The test set was not used during model optimization, checkpoint selection, or threshold selection.

The resulting performance was:

Table 6.2 – Final Independent Test Performance

Metric	Result
Accuracy	**93.33%**
Sensitivity	**82.00%**
Specificity	**99.00%**
PPV / Precision	**97.62%**
NPV	**91.67%**
F1-Score	**88.46%**
ROC-AUC	**0.9858**
Decision Threshold	**0.50**

The overall accuracy of **93.33%** indicates that 140 of the 150 test eyes were classified correctly. The model showed particularly strong specificity, reaching **99.00%**, which means that almost all Non-KC eyes were correctly recognized.

Sensitivity was **82.00%**, indicating that 41 of the 50 KC eyes were correctly detected. The remaining nine KC cases were classified as Non-KC.

The **PPV of 97.62%** indicates that when the model predicted Keratoconus, the prediction was correct in the large majority of cases. The NPV of **91.67%** indicates that most eyes predicted as Non-KC were actually Non-KC.

The difference between sensitivity and specificity is important. The model was highly effective at avoiding false-positive KC predictions, but it missed some KC cases. Therefore, the overall accuracy should not be considered in isolation; sensitivity and the confusion matrix provide important additional information about model behavior.

6.5 Confusion Matrix Analysis

The independent test confusion matrix provides a direct view of the four possible classification outcomes.

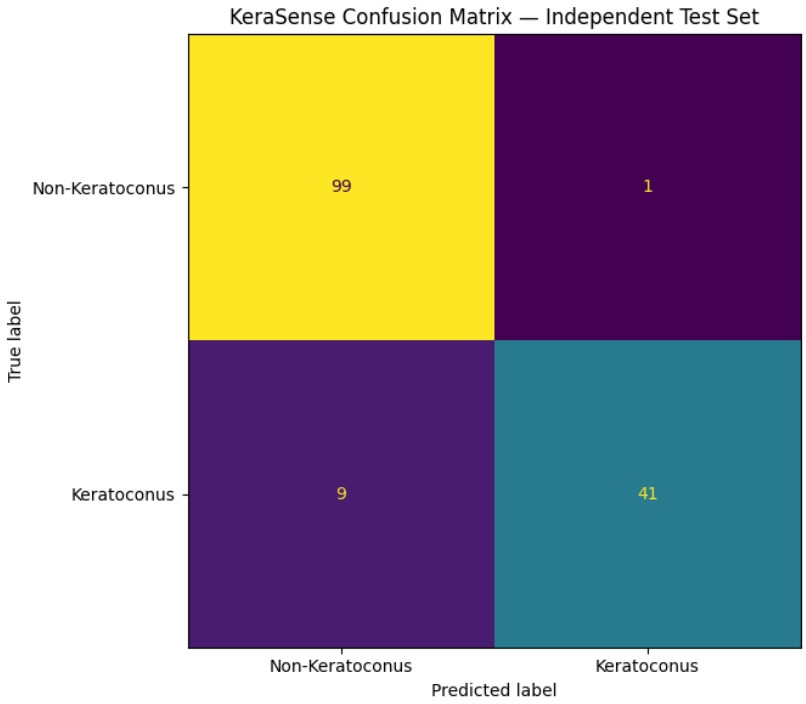


Figure 6.2: Confusion matrix of the final KeraSense model on the independent test set.

The matrix contains:

- **True Negatives (TN) = 99**
- **False Positives (FP) = 1**
- **False Negatives (FN) = 9**
- **True Positives (TP) = 41**

Executive Summary(1)

The model correctly classified **99 of the 100 Non-KC eyes**, producing only one false-positive prediction. This explains the very high specificity of 99%.

For the KC class, **41 of 50 eyes** were correctly detected. Nine KC eyes were incorrectly classified as Non-KC, resulting in the sensitivity of 82%.

This pattern indicates that the model has a strong tendency to correctly reject Non-KC cases but is less consistent when identifying every KC case. The false-negative group is therefore an important area for further investigation, particularly if the system is eventually intended for screening-oriented applications.

Table 6.3 – Numeric Confusion Matrix

Actual / Predicted	Non-KC	KC
Non-KC	99	1
KC	9	41

The confusion matrix also explains the reported accuracy:

$$Accuracy = \frac{99 + 41}{150}$$

$$Accuracy = 0.9333 = 93.33\%$$

6.6 ROC Curve and AUC Analysis

The Receiver Operating Characteristic (ROC) curve was generated using the continuous KC probability produced by the model. Unlike accuracy, which depends on a particular threshold, the ROC curve evaluates model discrimination across different probability thresholds.

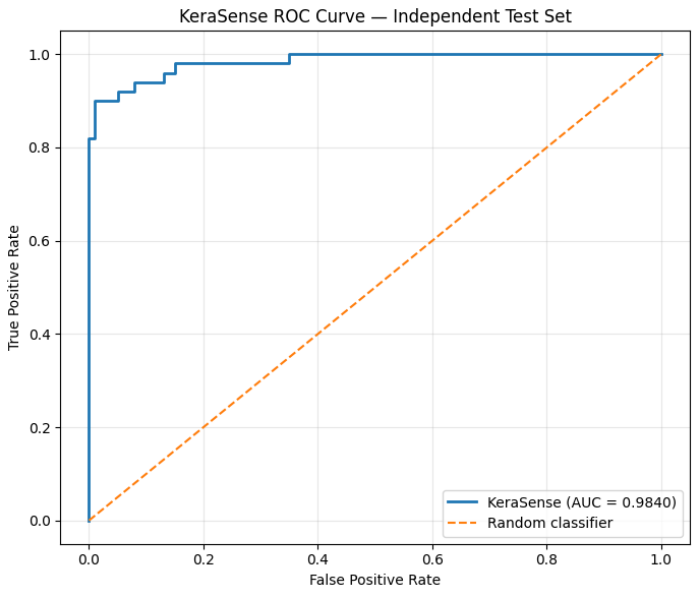


Figure 6.3: Receiver Operating Characteristic curve of KeraSense on the independent test set. The area under the curve is ROC-AUC = 0.9858.

The resulting **ROC-AUC was 0.9858**

An AUC close to 1 indicates that the model is generally able to assign higher KC probabilities to keratoconus eyes than to Non-KC eyes. The high AUC therefore demonstrates strong class-separation capability across probability thresholds.

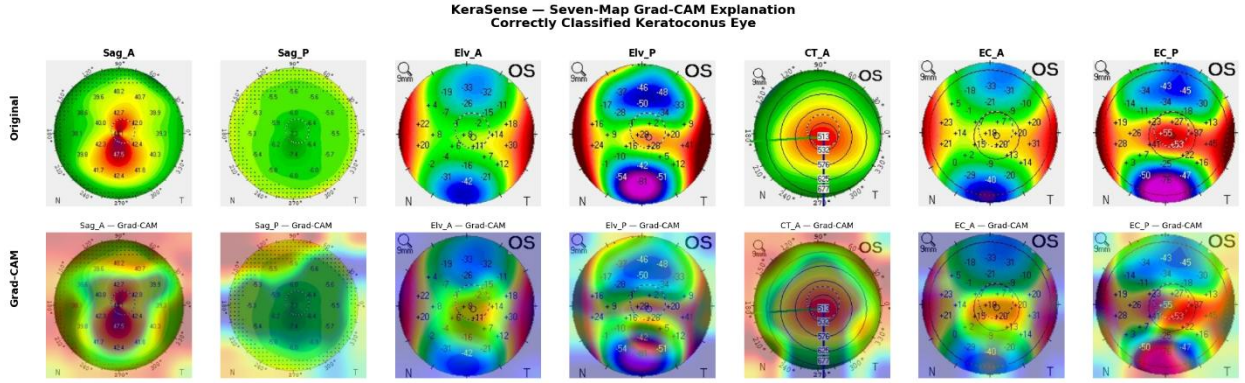
It is important, however, to distinguish **discrimination** from **threshold-based sensitivity**. The model's AUC is very high, while the sensitivity at the selected 0.50 threshold is 82%. This means that the model's probability ranking is strong overall, but the particular operating threshold does not identify every KC eye.

This observation is useful when interpreting the model: a high AUC does not automatically imply perfect sensitivity at every threshold.

6.7 Grad-CAM Explainability Results

. One of the important components of KeraSense is its ability to provide visual explanations for individual predictions. Grad-CAM was applied separately to each of the seven corneal maps processed by the EfficientNet-B0 backbone.

The resulting visualization allows the original map to be compared with the corresponding activation heatmap. Rather than producing a single explanation for the complete eye, the approach provides **seven map-specific explanations**



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Figure 6.4: Map-wise Grad-CAM visualization for a correctly classified keratoconus eye. The heatmaps indicate image regions contributing to the model prediction across the seven complementary corneal representations.

For a correctly classified KC eye, the Grad-CAM maps provide an opportunity to inspect whether the network concentrates on visually meaningful regions of the corneal representations. Areas associated with strong activation can be compared across curvature, elevation, thickness, and eccentricity maps.

The heatmaps should be interpreted as **model-attribution visualizations rather than anatomical segmentation**. A highlighted region indicates that the corresponding image features contributed to the prediction; it does not establish that the highlighted region is independently diagnostic of keratoconus.

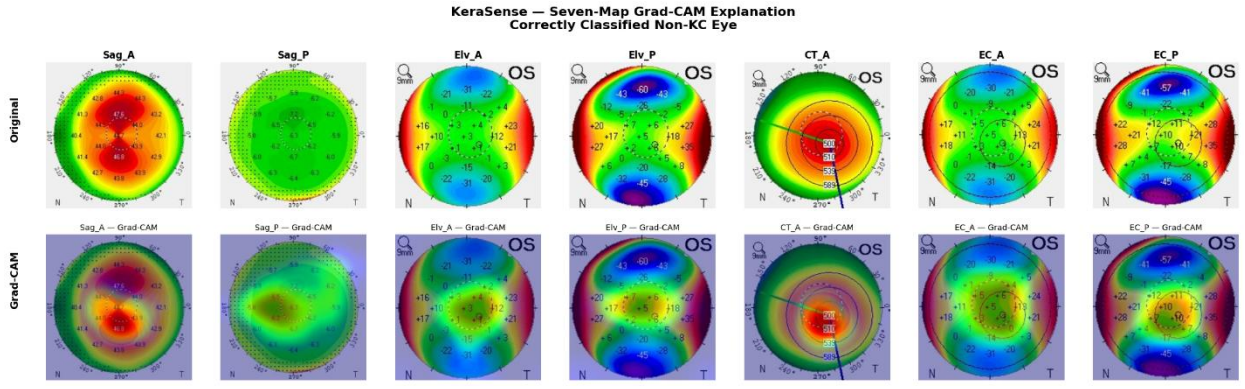


Figure 6.5: Map-wise Grad-CAM visualization for a correctly classified Non-Keratoconus eye. The figure illustrates the regions contributing to the model's Non-KC prediction across the seven input maps.

Comparing the KC and Non-KC cases provides a more useful interpretation than presenting a single heatmap. It allows the behavior of the network to be examined under two different classification outcomes.

Where the KC example shows concentrated activation in particular corneal regions, the Non-KC example can be examined to determine whether the model relies on substantially different spatial patterns.

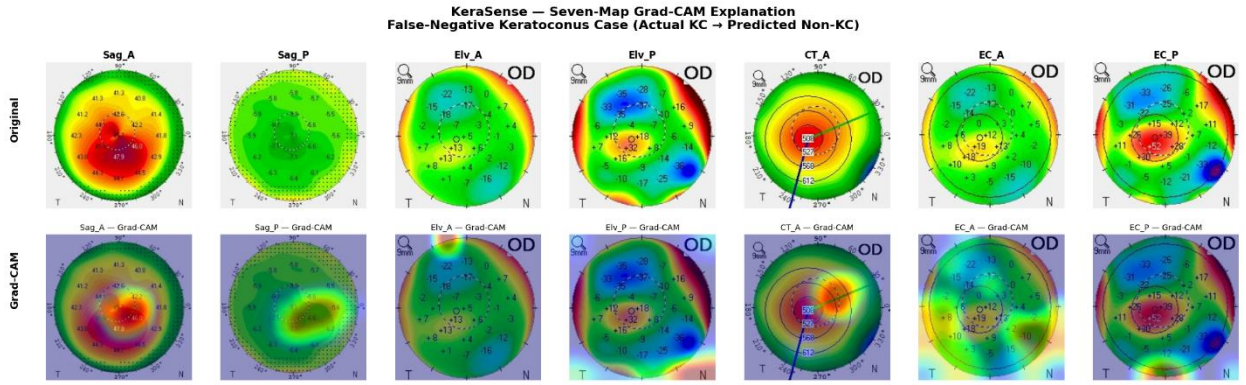


Figure 6.6: Map-wise Grad-CAM visualization for a false-negative keratoconus case. The visualization illustrates the model regions associated with an incorrect Non-KC prediction and highlights a limitation of the learned representation.

A false-negative example is particularly valuable because it demonstrates that the model does not perform perfectly on every KC case.

The independent test contained **nine false-negative KC predictions**. Executive Summary(1)

Examining one of these cases can help identify whether the model failed because the visual characteristics were subtle, distributed across multiple maps, or insufficiently represented by the learned features.

This type of analysis strengthens the report because it presents both successful predictions and model limitations rather than showing only favorable examples.

6.8 Prediction Examples

The KeraSense interface produces a probability for both classes in addition to the final classification. A representative KC prediction can therefore be reported using the actual interface output.

For example:

Prediction: Keratoconus

Keratoconus Probability: 91.27%

Non-Keratoconus Probability: 8.73%

Decision Threshold: 0.50

Table 6.4 – Representative KeraSense Prediction

Output	Value
Predicted class	Keratoconus
KC probability	91.27%
Non-KC probability	8.73%
Decision threshold	0.50
Number of maps analyzed	7

This example demonstrates how the model's probability output is converted into a final classification. Since the KC probability is greater than the fixed threshold of 0.50, the final prediction is Keratoconus.

The example should be treated as an illustration of the inference process rather than as evidence of overall model performance. The overall performance is determined from the complete independent test set.

6.9 Comparison with Previous Experiments

The final binary KeraSense model was developed after earlier experiments using the same general seven-map dataset. The earlier experiment attempted a **three-class classification problem involving Normal, Suspect, and Keratoconus**.

The earlier multiclass model achieved substantially lower independent-test performance, with test accuracy around **59.3%**. Its main difficulty was distinguishing Normal from Suspect eyes, while clearer Keratoconus cases were comparatively easier to identify. KeraSense: Multimodal Corneal I…

The final binary formulation instead combines Normal and Suspect into Non-KC and focuses on the KC versus Non-KC screening task.

Table 6.5 – Comparison of KeraSense Experiments

Experiment	Classification task	Test accuracy	Main observation
Earlier KeraSense experiment	Normal / Suspect / KC	~59.3%	Stronger difficulty separating Normal and Suspect
Final KeraSense	**Non-KC / KC**	**93.33%**	Strong binary discrimination with high specificity

The difference between the two experiments demonstrates the effect of task formulation. The three-class problem requires the model to distinguish among three visually related categories, including the intermediate Suspect class. The binary experiment instead combines Normal and Suspect into one group and concentrates on detecting Keratoconus.

The final model therefore provides a much clearer separation between the target classes on the independent test set.

However, the improvement should not be interpreted simply as evidence that the final architecture is universally superior. **The classification task itself changed**, and therefore the two experiments are not directly equivalent. The comparison is useful primarily for understanding how the binary formulation affected the behavior of the KeraSense system.

6.10 Overall Results Discussion

The final KeraSense experiment demonstrates that the seven-map representation can provide a useful basis for binary keratoconus classification. The model achieved **93.33% accuracy**, **99.00% specificity**, **82.00% sensitivity**, and **0.9904 ROC-AUC** on 150 independent test eyes. Executive Summary(1)

The strongest result is the model's specificity. Only one Non-KC eye was incorrectly classified as KC. At the same time, nine KC eyes were classified as Non-KC, showing that sensitivity remains an important limitation.

The very high ROC-AUC suggests that the model has strong overall discrimination, while the lower sensitivity at the selected threshold demonstrates that probability ranking and threshold-based classification are not identical measures of performance.

The Grad-CAM analysis adds an interpretability component by allowing the contribution of individual corneal maps to be examined. This is especially relevant to a multimodal system because different maps may provide different visual evidence for the same prediction.

The final results therefore support the feasibility of the KeraSense approach as an **AI-assisted research prototype for keratoconus screening**, while the false-negative cases and absence of external validation indicate that further evaluation would be required before any clinical application.

7 DISCUSSION

This chapter interprets the experimental findings of KeraSense and discusses their significance rather than repeating the numerical results presented in Chapter 6. The discussion focuses on what the observed performance suggests about multimodal corneal analysis, the binary screening formulation, model behavior, explainability, and the practical relevance of the proposed system. KeraSense is considered a **research prototype for AI-assisted screening**, rather than a clinically validated diagnostic system. Executive Summary(1)

7.1 Interpretation of the Results

The results indicate that KeraSense was able to learn discriminative patterns from the seven corneal representations when they were considered jointly. The strong performance on the independent test set suggests that the model learned features that were sufficiently consistent between the development and test portions of the dataset. This is particularly relevant because each prediction was based on the complete set of seven maps from the same eye rather than on an isolated image.

The training behavior also provides an important interpretation of the model development process. The network reached high training performance relatively quickly, while validation performance remained strong until the selected checkpoint. The subsequent increase in validation loss indicates that continuing training would increasingly favor patterns specific to the training data. Selecting the checkpoint according to validation loss therefore helped prevent the final model from simply using the training set as its source of evidence. The use of an untouched independent test set further separates model development from final evaluation. Executive Summary(1)

The test results show a particularly strong ability to reject Non-KC cases. At the same time, the remaining missed KC cases demonstrate that good overall classification does not mean that every keratoconus eye presents an equally recognizable pattern to the model. This distinction is important in ophthalmic screening, where subtle or atypical disease patterns can be more difficult to separate from non-diseased or suspect cases.

Therefore, the main interpretation is not simply that KeraSense produces a high classification score. Rather, the experiment demonstrates the **feasibility of learning a shared representation from multiple corneal maps for binary KC screening**, while also revealing the need for stronger validation and improved handling of difficult cases.

7.2 Effect of Multimodal Corneal Map Fusion

One of the central ideas behind KeraSense is that keratoconus is not represented by a single structural property. Different corneal maps describe different aspects of the same eye. Curvature maps provide information about surface steepening, elevation maps describe deviations of the corneal surfaces, pachymetry represents tissue thickness, and eccentricity maps capture changes in corneal shape. Considering these representations together provides a broader description of corneal structure. Executive Summary(1)

KeraSense implements this concept by processing each of the seven maps through a shared EfficientNet-B0 feature extractor and subsequently concatenating the resulting feature vectors. Consequently, the classifier receives information from all seven representations before making the final decision. This is different from relying on a single curvature or elevation map and allows complementary patterns to contribute to the prediction.

An important aspect of this design is that the model does not require the seven maps to be manually converted into handcrafted numerical indices before classification. Instead, the feature extractor learns visual representations directly from the map images. The fused representation can therefore contain information from multiple structural characteristics simultaneously.

This approach is consistent with previous research showing the usefulness of multiple corneal maps for keratoconus analysis. In particular, Al-Timemy et al. used seven corneal maps with EfficientNet-based feature extraction and feature fusion. KeraSense follows a related multimodal principle but uses a **shared EfficientNet-B0 backbone**, providing a more unified feature-extraction structure. Executive Summary(1)

However, the present experiment does **not** establish that every individual map contributes equally or that fusion always outperforms every possible single-map model. An explicit ablation study would be required to make that claim. The appropriate conclusion is that the fused seven-map representation provides a feasible and effective basis for the present binary classification task.

7.3 Performance of the Binary Screening Approach

The decision to formulate the final task as **Keratoconus versus Non-Keratoconus** simplifies the clinical question being addressed by the model. Normal and Suspect cases were grouped into the Non-KC category, while confirmed Keratoconus cases formed the positive class. This allows the model to focus directly on distinguishing eyes requiring greater attention from those that do not belong to the KC class within the selected dataset.

This formulation also avoids treating Normal and Suspect as completely independent disease categories when the primary objective is KC screening. A screening system can first determine whether an eye appears consistent with KC and subsequently allow clinical assessment or additional testing to determine its exact status.

The results suggest that this binary formulation is learnable with the available data. However, grouping Suspect eyes with Normal cases also introduces an important limitation. Suspect eyes can contain subtle structural abnormalities and may not be equivalent to clearly normal eyes. Consequently, the Non-KC class is internally heterogeneous. This can make some samples more difficult to classify and may partly explain why the model does not detect every KC case.

Thus, the binary approach should be interpreted as a **screening-oriented formulation**, rather than as a complete clinical categorization of corneal disease. A future system could extend this framework to distinguish Normal, Suspect, and Keratoconus separately or introduce additional risk categories.

7.4 Interpretation of Sensitivity and Specificity

The difference between sensitivity and specificity provides one of the most important insights from the evaluation.

The model demonstrated **very strong specificity**, meaning that only a small number of Non-KC eyes were incorrectly labelled as Keratoconus. From a screening-support perspective, this indicates that the model was conservative in assigning the positive KC class. A positive prediction therefore carried a high likelihood of representing a true KC case within the evaluated dataset.

Sensitivity was comparatively lower. The false-negative cases show that some keratoconus eyes were not sufficiently separated from the Non-KC representation learned by the model. This is clinically important because a missed KC case is more consequential than an unnecessary referral in many screening scenarios.

The threshold selection also influences this trade-off. KeraSense uses a KC probability threshold of 0.50 selected using the validation data. In general, reducing this threshold would make the system more willing to classify an eye as KC, potentially increasing sensitivity at the expense of specificity. Conversely, increasing the threshold would make positive predictions more conservative.

Therefore, the selected threshold should not be interpreted as a universally optimal clinical threshold. It is the operating point established for this experiment. A larger clinical dataset, particularly with more subtle and early-stage KC cases, would be required to determine an operating threshold appropriate for real-world screening.

7.5 Role of Grad-CAM Explainability

The Grad-CAM analysis adds an interpretability component to the numerical prediction. Instead of reporting only whether an eye was classified as KC or Non-KC, KeraSense provides map-specific visualizations showing regions that contributed to the selected class prediction.

This is particularly relevant for multimodal corneal analysis because the seven maps contain different structural information. Examining the Grad-CAM outputs across the maps allows the behavior of the model to be considered in relation to the different representations rather than treating the prediction as a completely opaque result.

The correctly classified KC example provides evidence that the model can focus on visually meaningful regions associated with abnormal corneal structure. The correctly classified Non-KC example provides a contrasting case in which the learned activation pattern does not strongly support the KC class. The false-negative example is particularly valuable because it demonstrates that an incorrect prediction can also be investigated rather than simply recorded as a classification error.

Nevertheless, Grad-CAM should **not** be interpreted as a clinical proof of causality. A highlighted region indicates that the corresponding features influenced the network output; it does not establish that the highlighted region represents a medically confirmed keratoconus lesion. This distinction is important when using explainability techniques in medical AI. The Grad-CAM results should therefore be considered a tool for model inspection and hypothesis generation rather than a replacement for clinical interpretation. The use of Grad-CAM in KeraSense follows the broader use of activation-based explanations for deep-learning analysis of corneal maps.

7.6 Comparison with Existing Research

The performance of KeraSense is consistent with the broader direction of research in AI-assisted keratoconus detection. Previous studies have demonstrated that deep-learning models can learn useful discriminative representations from corneal curvature, elevation, and tomography images. For example, Kuo et al. demonstrated strong performance using CNN-based analysis of corneal curvature maps, while Fassbind et al. investigated deep convolutional models using corneal topography representations. Executive Summary(1)

The most closely related work in terms of input structure is the study by Al-Timemy et al., which used seven corneal maps and EfficientNet-based feature extraction. This provides an important reference point for KeraSense because both approaches recognize that multiple map representations can provide complementary information. KeraSense differs principally in its use of a shared EfficientNet-B0 feature extractor followed by direct feature concatenation and an MLP classification head, together with map-wise Grad-CAM visualization.

At the same time, direct numerical comparison between studies should be treated cautiously. Differences in datasets, class definitions, imaging devices, sample sizes, preprocessing, train-test protocols, and evaluation criteria can substantially affect reported performance. Therefore, a higher or lower percentage than another published study does not by itself establish superiority. The more meaningful contribution of KeraSense is its **end-to-end integration of seven-map input, shared deep feature extraction, multimodal fusion, binary screening, independent testing, and visual explanation within one reproducible prototype**. This places the work within the existing research direction while retaining a clearly defined experimental scope.

7.7 Practical Significance of KeraSense

The practical significance of KeraSense lies primarily in its potential as an **AI-assisted screening and decision-support prototype**. A clinician or researcher can provide the seven corneal maps from an eye and receive a binary prediction together with a probability score and map-specific visual explanations. This creates a workflow in which automated pattern recognition can be combined with visual inspection rather than replacing it.

The multimodal design is particularly useful because it mirrors the fact that corneal assessment involves several complementary representations. Instead of requiring the user to interpret each map independently before forming an overall impression, KeraSense provides a unified computational assessment of the seven-map set.

However, the current results should not be interpreted as evidence that KeraSense is ready for clinical deployment. The evaluation is based on retrospective data from a fixed dataset, and the system has not undergone prospective clinical testing or external validation across different patient populations and imaging devices. The available dataset is also relatively small for a deep-learning application, and the model does not incorporate additional clinical information such as patient history, refraction, or biomechanical measurements. Executive Summary(1)

Consequently, the most appropriate practical role for the current system is as a **research and decision-support prototype**. Its value is in demonstrating how multiple corneal representations can be combined with deep learning and explainability in a single workflow. Future validation using larger, multicentre datasets and different imaging systems would be necessary before assessing its suitability for routine clinical use.

Overall, KeraSense demonstrates that multimodal corneal image analysis can provide a useful foundation for automated keratoconus screening. Its strongest contribution is not simply the final classification score, but the integration of **multimap analysis, learned feature fusion, independent evaluation, and visual explanation** into a single framework.

8 LIMITATIONS AND FUTURE SCOPE

8.1

Limitation	Description
Limited dataset size	**The dataset contains a relatively limited number of eyes for training and evaluating a deep-learning model. This restricts the extent to which the results can be generalized.**
Seven maps are not independent samples	**Each eye is represented by seven complementary corneal maps. These maps originate from the same eye and therefore should not be treated as seven independent patient samples.**
Image-only input	**The current KeraSense model uses only the seven corneal map images. Clinical information such as patient history, refraction, and additional biomechanical measurements is not included.**
Limited multimodal fusion strategy	**The current architecture combines the extracted features through direct concatenation. It does not use an explicit attention mechanism to dynamically determine the importance of individual corneal maps.**
Potential for improved feature representation	**More advanced fusion approaches could potentially learn stronger relationships between the complementary corneal maps and improve the representation used for classification.**

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8.2 Dataset and Generalization Limitations

The size and composition of the dataset limit the extent to which the results can be generalized. The development data contain **423 eyes**, while the independent test set contains **150 eyes**. Although the test set was kept separate from model development, it comes from the **same overall dataset source** as the development data. Therefore, it should not be considered a true external validation cohort.

This distinction is important. A model may perform well when the training and test data originate from the same population, imaging system, and acquisition conditions, while showing different performance on images collected using another device or from another patient population.

External validation was not performed in the current study. Consequently, it is not possible to determine from the present experiment how well KeraSense would perform across different hospitals, ethnic populations, imaging devices, acquisition protocols, or disease distributions.

The limited number of eyes also restricts the reliability of conclusions about uncommon or difficult cases. Increasing the number of labelled eyes, particularly challenging Suspect and early-KC cases, would provide a stronger basis for evaluating generalization.

8.3 Clinical Validation Requirements

KeraSense has **not undergone clinical validation** and should not be considered a clinically approved diagnostic system. The current experiment evaluates the model computationally using a fixed research dataset. It does not establish how the system would perform when integrated into an actual ophthalmology workflow.

Before clinical use could be considered, the model would require evaluation on independent patient cohorts, preferably collected from multiple clinical centres. Prospective studies would

also be necessary to determine whether the predictions provide useful information to clinicians under realistic conditions.

The system should therefore be viewed as an **AI-assisted research and decision-support prototype**. Its predictions are intended to support further examination rather than replace ophthalmological assessment.

In addition, regulatory, ethical, and workflow considerations would need to be addressed before any clinical deployment.

8.4 Future Model Improvements

Several improvements could be investigated in future versions of KeraSense.

First, a larger and more diverse dataset could improve the stability of the learned representations. Additional data augmentation and carefully designed sampling strategies could also help the model handle difficult cases.

Second, the current feature-fusion mechanism could be extended. Instead of simply concatenating the seven feature vectors, an **attention-based fusion mechanism** could learn which corneal maps are more informative for an individual eye. This could allow the network to give greater weight to the maps that contain the strongest evidence for a particular prediction.

Third, additional information could be incorporated into the model. Corneal biomechanical measurements, clinical measurements, refractive information, or patient-level characteristics could complement the image-based features. Such an extension would provide a broader multimodal representation than the current image-only system.

The classification task could also be extended beyond the current binary formulation. Future work could investigate separate **Normal, Suspect, and Keratoconus** classification or more detailed severity categories. This would be particularly useful for studying borderline and early-stage cases.

Finally, systematic ablation experiments could be performed to determine the individual and combined contribution of the seven maps. Removing one map at a time or comparing different fusion strategies would provide stronger evidence about which representations contribute most to the final prediction.

8.5 External and Multicenter Validation

External validation is one of the most important next steps for KeraSense. The current independent test set provides a useful evaluation because it was not used during model training, but it still originates from the same overall dataset source. It therefore does not measure performance under a genuinely different data distribution.

A stronger validation strategy would involve testing the trained model on datasets obtained from **different institutions and imaging systems**. Multicenter validation could determine whether the learned features remain useful when acquisition conditions and patient populations change.

Such testing could also reveal whether the model has learned general corneal patterns or has partially relied on dataset-specific characteristics. Evaluation across different devices would be particularly valuable because corneal maps can differ in appearance and numerical representation depending on the imaging system.

A future study should therefore follow a structure such as:

Development dataset → internal validation → independent external dataset → multicenter evaluation

Only after such evaluation could the generalization and robustness of KeraSense be assessed with greater confidence.

8.6 Longitudinal Keratoconus Progression Analysis

Longitudinal progression prediction was **not implemented in the current KeraSense system**. The present study uses cross-sectional eye data and focuses on identifying keratoconus from the available seven-map representation. It does not contain the repeated patient visits and validated progression labels required to determine whether an eye will progress over time.

Therefore, the term **deformation analysis** in this project should be understood carefully. It refers to the analysis of **structural patterns represented across complementary corneal maps**, such as curvature, elevation, thickness, and eccentricity. It does **not** represent a validated biomechanical deformation measurement or a direct measurement of how the cornea physically deforms over time.

Longitudinal analysis would require repeated examinations of the same eye at multiple time points. A possible future framework could analyse:

Visit 1 → Visit 2 → Visit 3 → structural changes → progression assessment

With suitable longitudinal data, future versions of KeraSense could investigate whether changes in the seven-map representations are associated with disease progression. Temporal models, recurrent architectures, transformers, or longitudinal feature-analysis methods could then be explored.

However, progression labels should come from clinically established criteria rather than being artificially created from the current cross-sectional dataset. This ensures that any future progression model represents a meaningful clinical outcome.

Overall Future Direction

The future development of KeraSense should therefore move in three main directions: **more diverse data, stronger clinical validation, and richer multimodal modelling**. The existing system establishes a foundation for seven-map corneal analysis, but larger external datasets and longitudinal clinical information are needed to determine whether the learned representations remain reliable beyond the current experimental setting.

The present work should consequently be viewed as a **feasibility study and research prototype**, rather than as a clinically validated diagnostic or progression-prediction system. This distinction keeps the claims of the project aligned with what was actually implemented and evaluated.

9 CONCLUSION

9.1 Summary of the Developed System

KeraSense was developed as a multimodal deep-learning framework for automated keratoconus screening using seven complementary corneal maps: **Sag_A, Sag_P, Elv_A, Elv_P, CT_A, EC_A, and EC_P**. The system applies a shared **EfficientNet-B0** feature extractor to the seven maps, combines the extracted features, and performs binary classification into **Keratoconus (KC)** and **Non-Keratoconus (Non-KC)**. A Grad-CAM-based explanation module was also incorporated to visualize regions contributing to individual predictions.

The complete workflow includes image preprocessing, eye-level data organization, model training, validation-based model selection, independent testing, and visual interpretation of predictions.

9.2 Major Findings

The experimental results demonstrate that the proposed approach can effectively learn discriminative information from the combined seven-map representation. The final model

achieved strong performance on the independent test set, particularly in identifying Non-KC eyes, while also detecting the majority of KC cases.

The Grad-CAM analysis further provided a way to inspect the model's predictions across the different corneal representations. Correctly classified cases demonstrated meaningful activation patterns, while the false-negative case highlighted that difficult KC patterns can still be missed.

These findings support the feasibility of combining **multimap corneal analysis, deep feature extraction, feature fusion, and explainability** within a single keratoconus screening framework.

9.3 Overall Conclusion

KeraSense demonstrates the potential of multimodal deep learning for automated analysis of keratoconus from complementary corneal maps. The study shows that combining information from multiple representations can provide a useful basis for KC screening while Grad-CAM can improve the interpretability of the resulting predictions.

However, the system remains a **research prototype rather than a clinically validated diagnostic tool**. Larger datasets, external and multicenter validation, and prospective clinical evaluation are required before its effectiveness in real-world clinical settings can be established.

Overall, the work provides a practical foundation for further development of **multimodal, explainable AI systems for corneal disease analysis**.

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